

NETWORK TROUBLESHOOTING AND DHCP/DNS CONFIGURATION

IN A VIRTUAL LAN ENVIRONMENT

Project Type: Networking / System Administration Project

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Role:
Network Administrator (Project Implementation)

Project Overview

This project demonstrates the configuration of DHCP and DNS services using Windows Server 2022 and Ubuntu client machines in a virtual LAN environment using VMware Workstation.

The project also includes network troubleshooting techniques such as ping, nslookup, traceroute, and IP configuration verification.

Technologies Used

- Windows Server 2022
 - Ubuntu 24.04
 - VMware Workstation
 - DHCP Server Configuration
 - DNS Server Configuration
 - Network Testing & Troubleshooting Tools
-

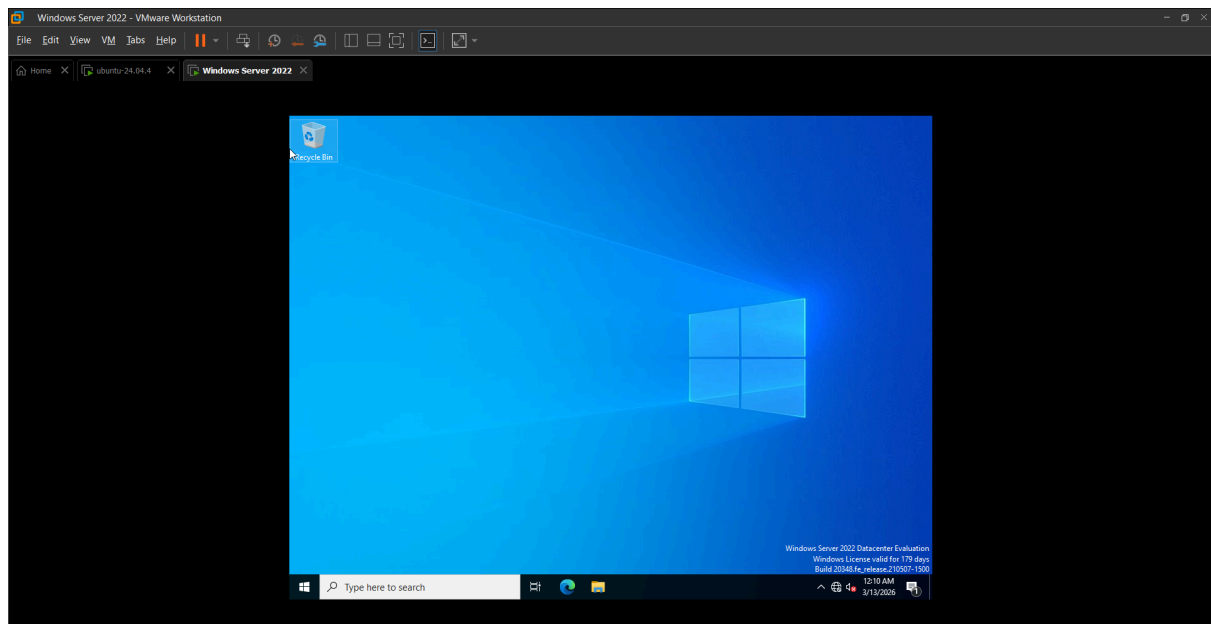
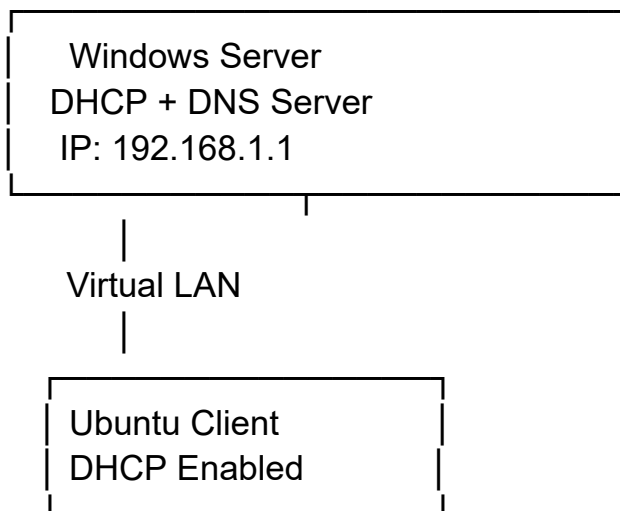
Key Skills Demonstrated

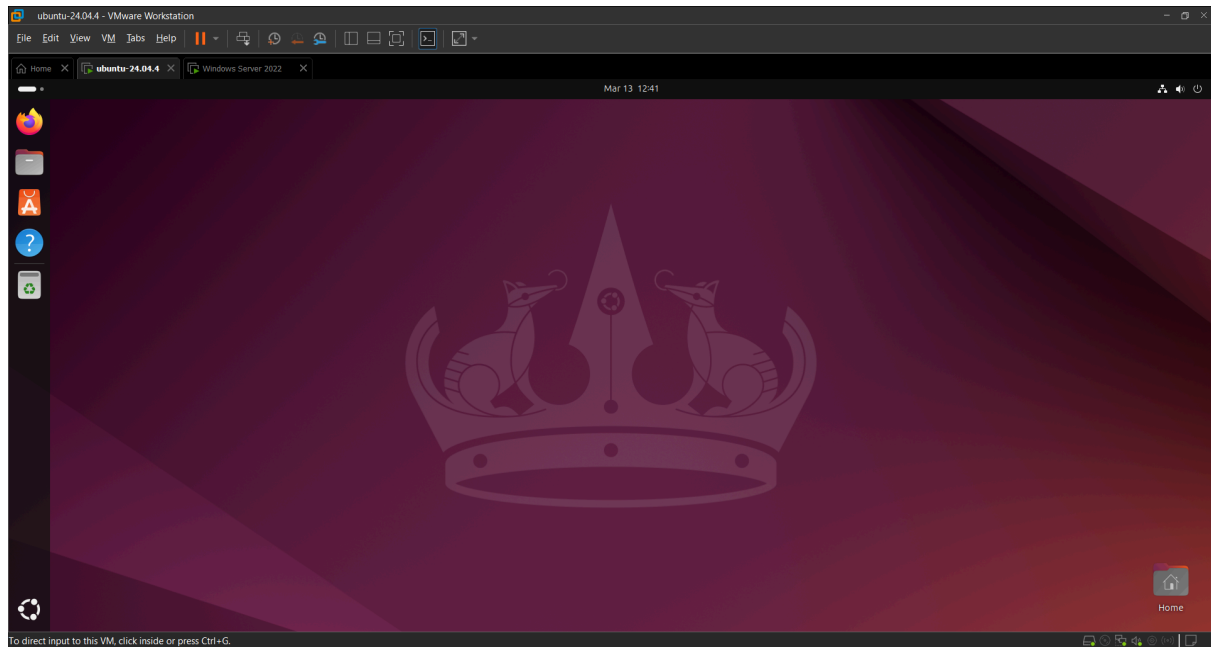
- Network Configuration
 - DHCP and DNS Setup
 - Virtual Network Management
 - Linux & Windows Server Administration
 - Network Troubleshooting
-

Project Year: 2026

Network Troubleshooting and DHCP/DNS Configuration in a Virtual LAN Environment

Network Architecture





Step 2: Configuring a Static IP Address in Windows Server (Detailed Explanation)

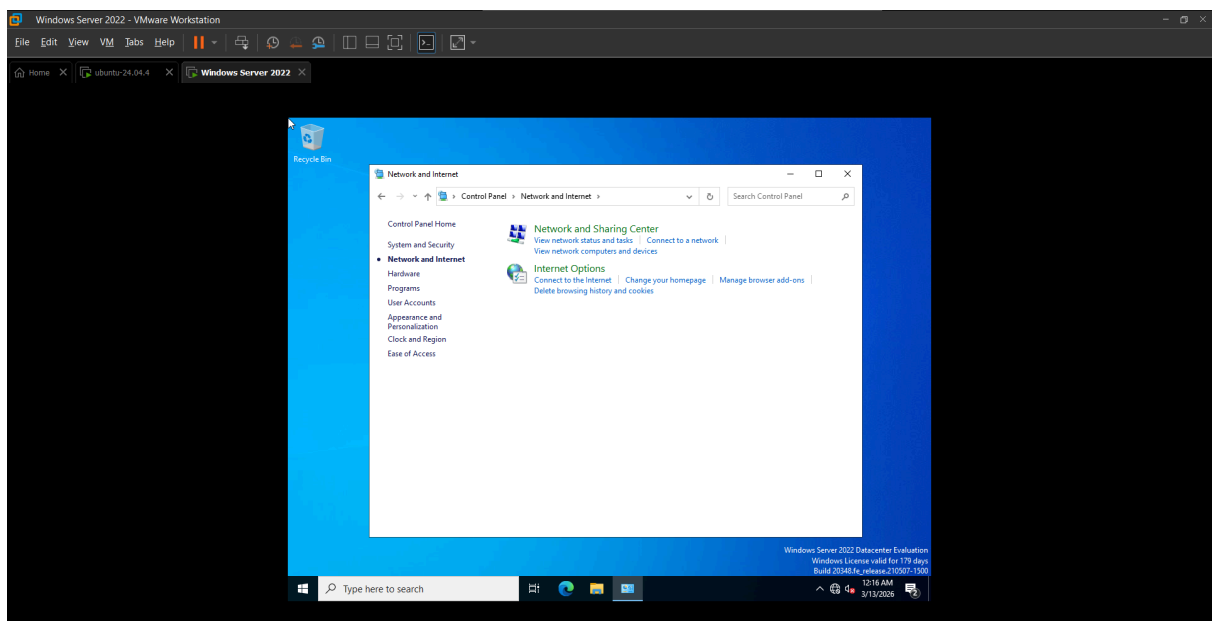
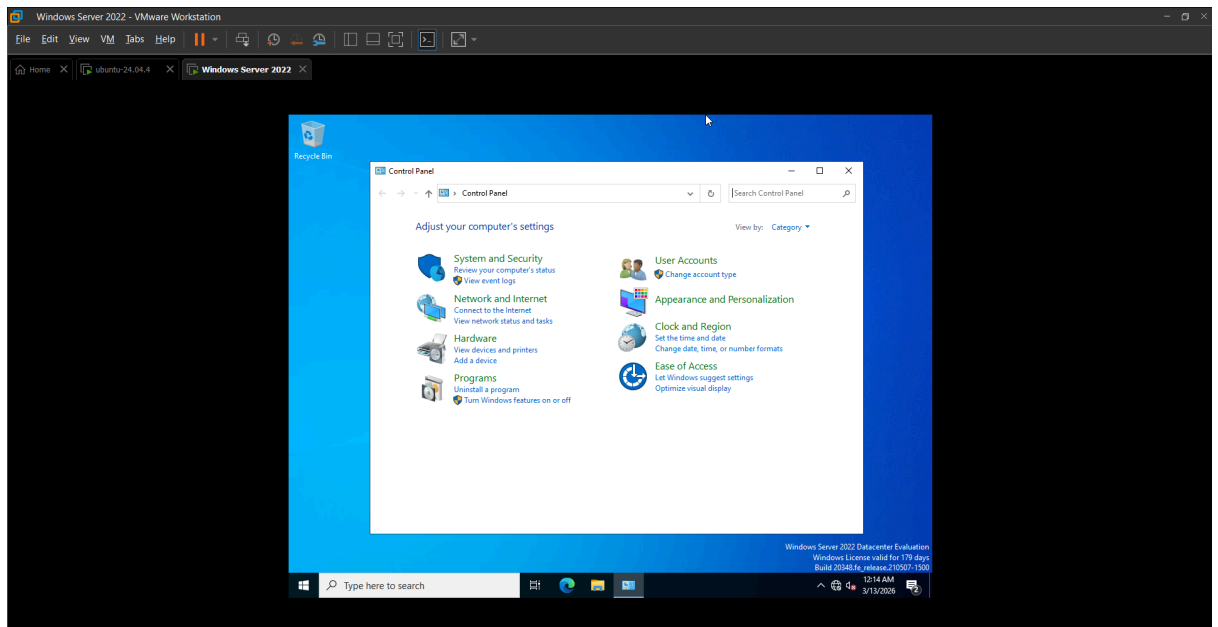
The purpose of this step is to assign a **fixed (static) IP address** to the Windows Server. Services such as **DHCP and DNS must always be available at the same IP address**, therefore a static IP is required. If the server's IP address changes, client machines may not be able to locate the server properly.

1 Opening Network Settings

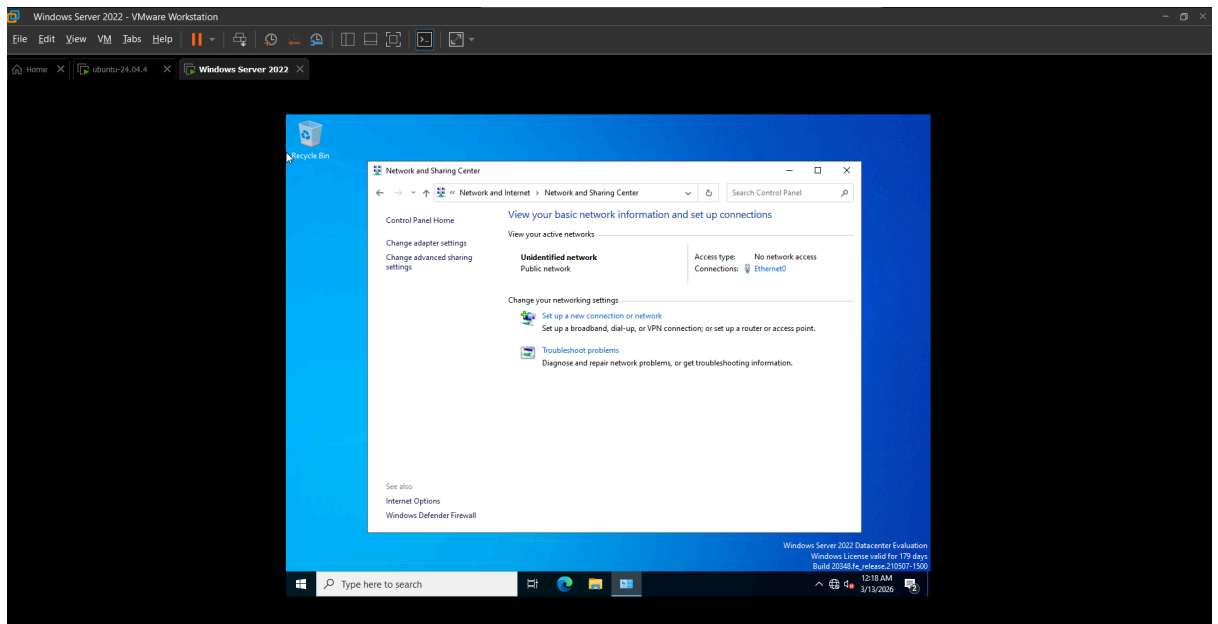
On the Windows Server:

1. Open **Control Panel**
2. Click on **Network and Internet**

3. Open **Network and Sharing Center**



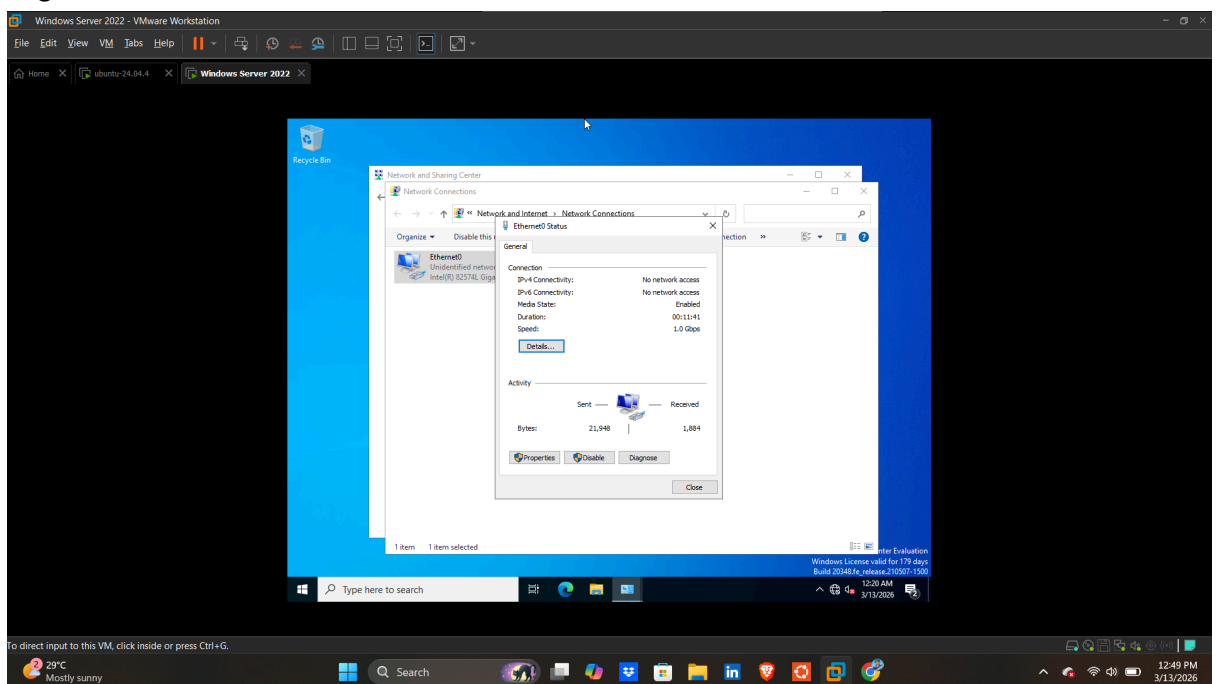
4. On the left side, click **Change adapter settings**



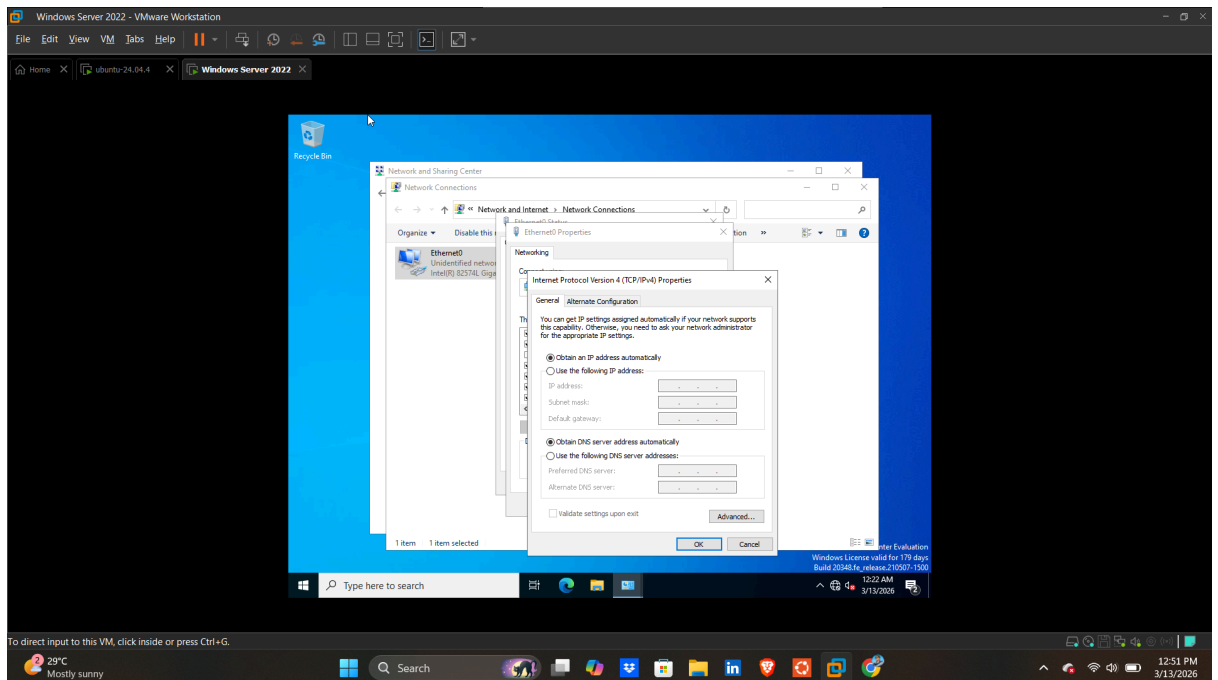
You will now see the **Ethernet network** adapter.

2 Opening Ethernet Properties

1. Right-click on **Ethernet**



2. Select Properties



A new window will appear showing a list of **networking components**.

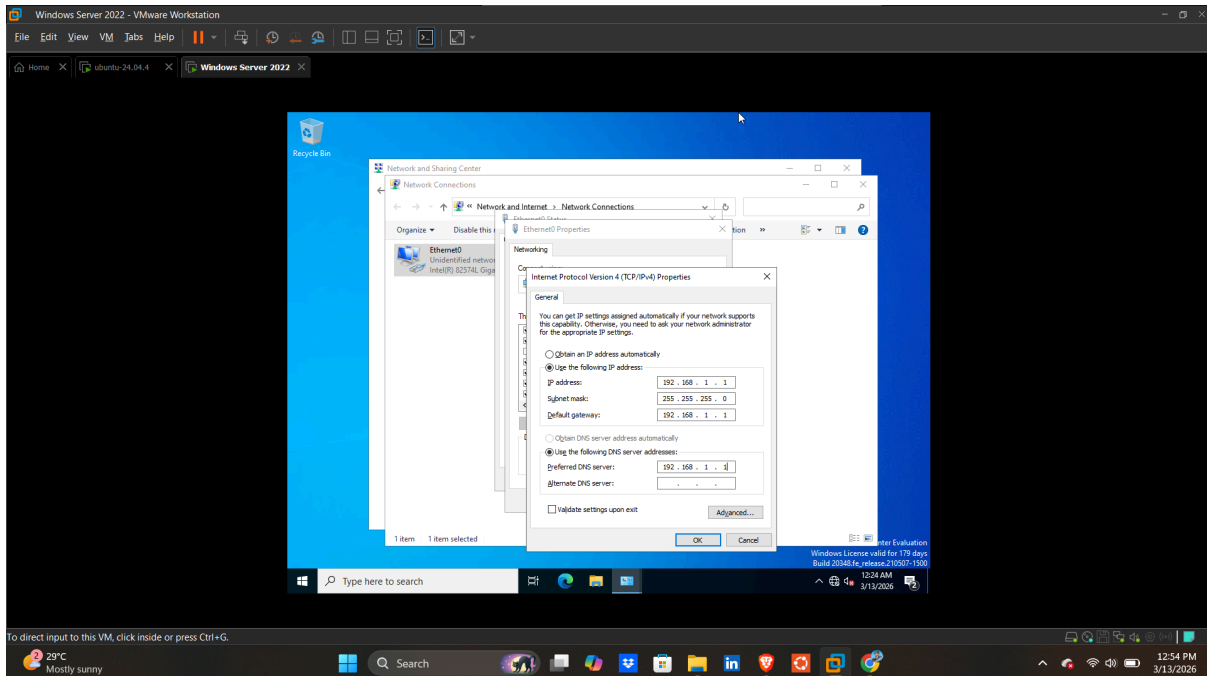
4 Setting the Static IP Address

Select **Use the following IP address** and enter the following values:

Setting	Value
IP Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

Preferred
DNS

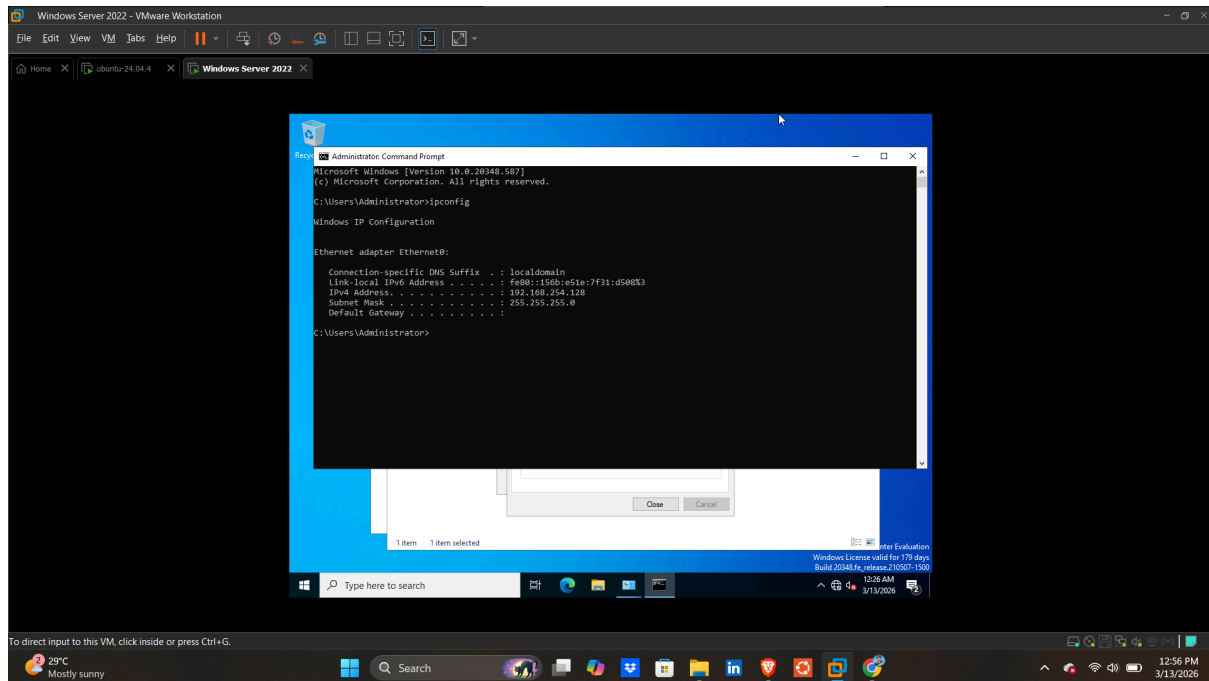
192.168.1.1



5 Verifying the Configuration

Open **Command Prompt** and run the following command:

ipconfig



Window server static IP setting success.

Step 3: Installing the DHCP Server Role (Windows Server)

The purpose of this step is to **install the DHCP service on Windows Server**, allowing the server to automatically assign IP addresses to client machines.

DHCP stands for **Dynamic Host Configuration Protocol**.

It automatically provides network devices with configuration details such as **IP address, subnet mask, default gateway, and DNS server**.

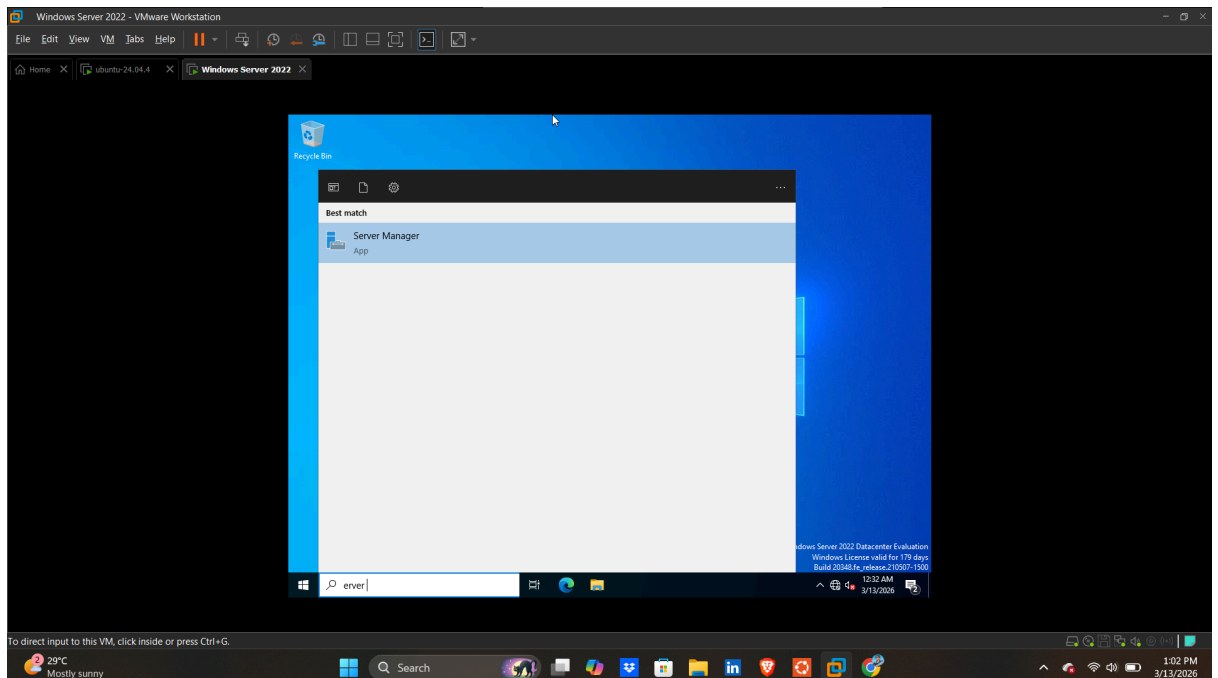
1 Opening Server Manager

When Windows Server starts, **Server Manager** usually opens automatically.

If it does not open:

1. Click the **Start Menu**

2. Search for **Server Manager**



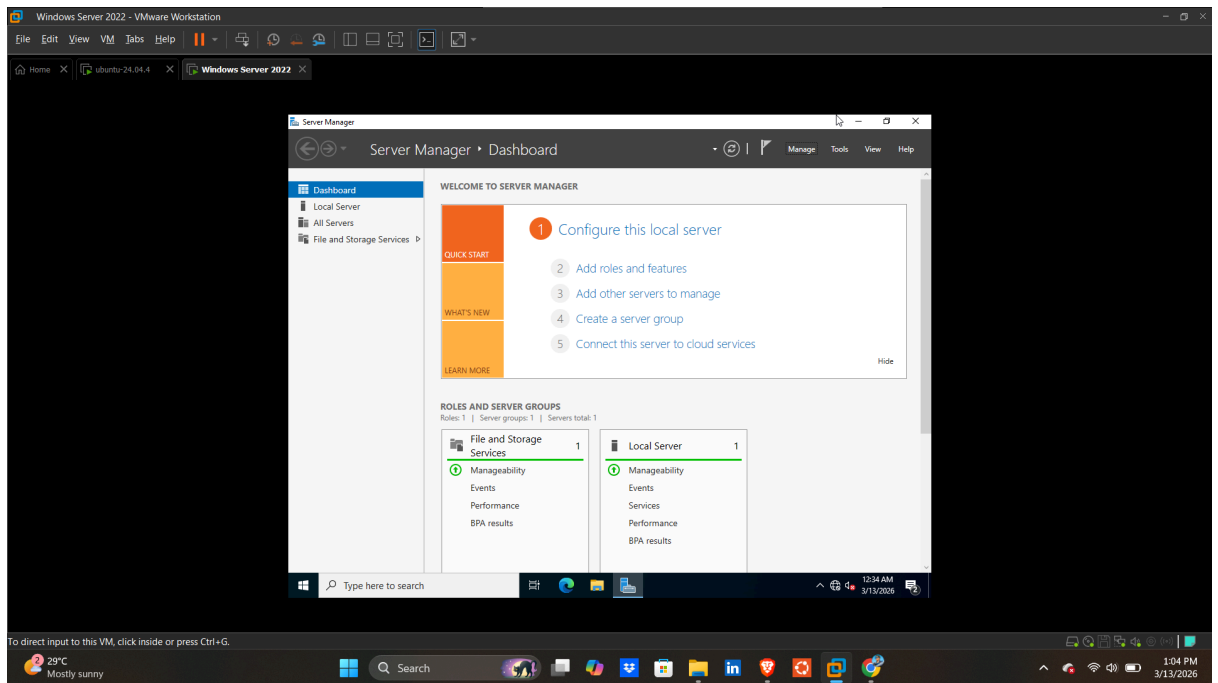
3. Open it

Server Manager is the **main management dashboard** of Windows Server where administrators can install and manage server roles and services.

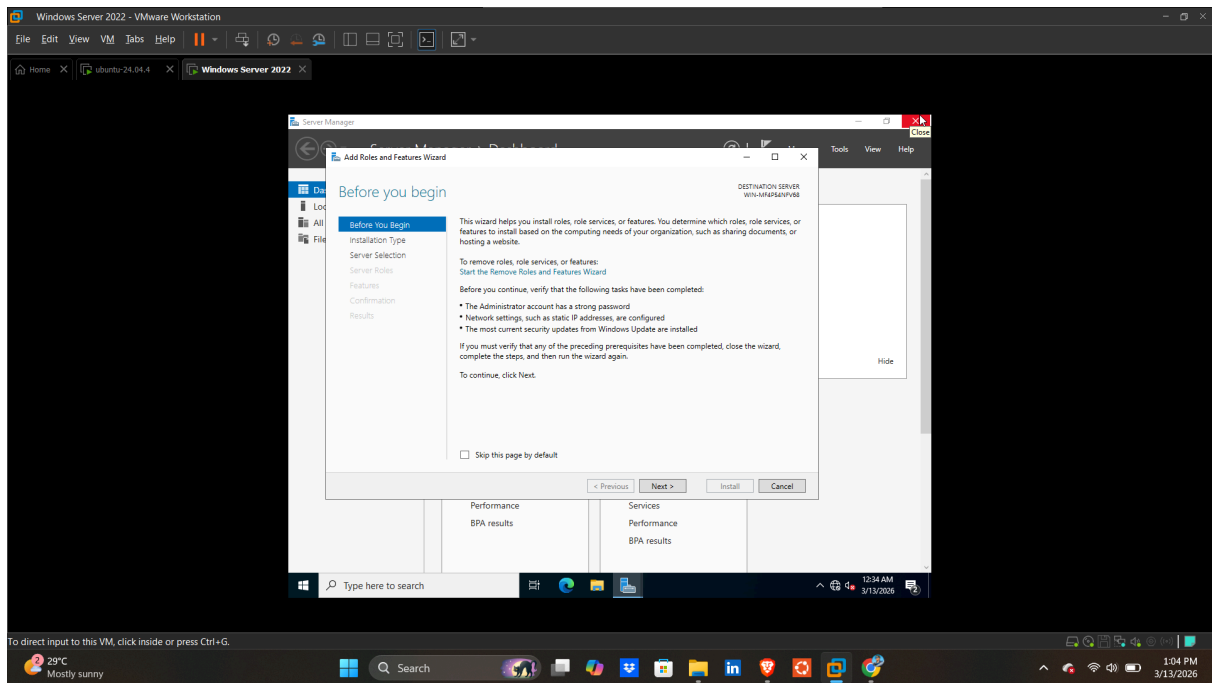
2 Starting the Add Roles and Features Wizard

In **Server Manager**:

1. Click **Manage** in the top menu



2. Select **Add Roles and Features**

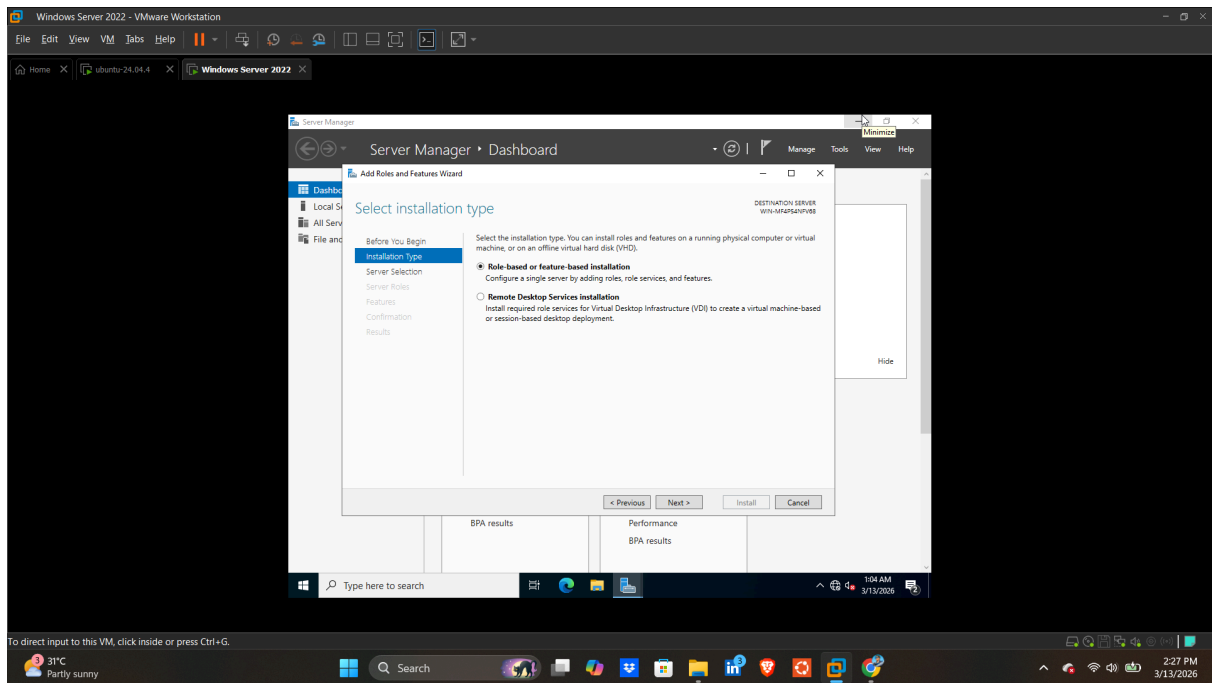


A wizard will open that is used to install **new roles and services** on the server.

3 Selecting the Installation Type

In the wizard:

1. Select **Role-based or feature-based installation**



2. Click **Next**

This option is used when installing specific server roles such as **DHCP**, **DNS**, or **Web Server (IIS)**.

4 Selecting the Server

On the next screen:

1. Select **Select a server from the server pool**

Usually, only one server will appear.

Example:

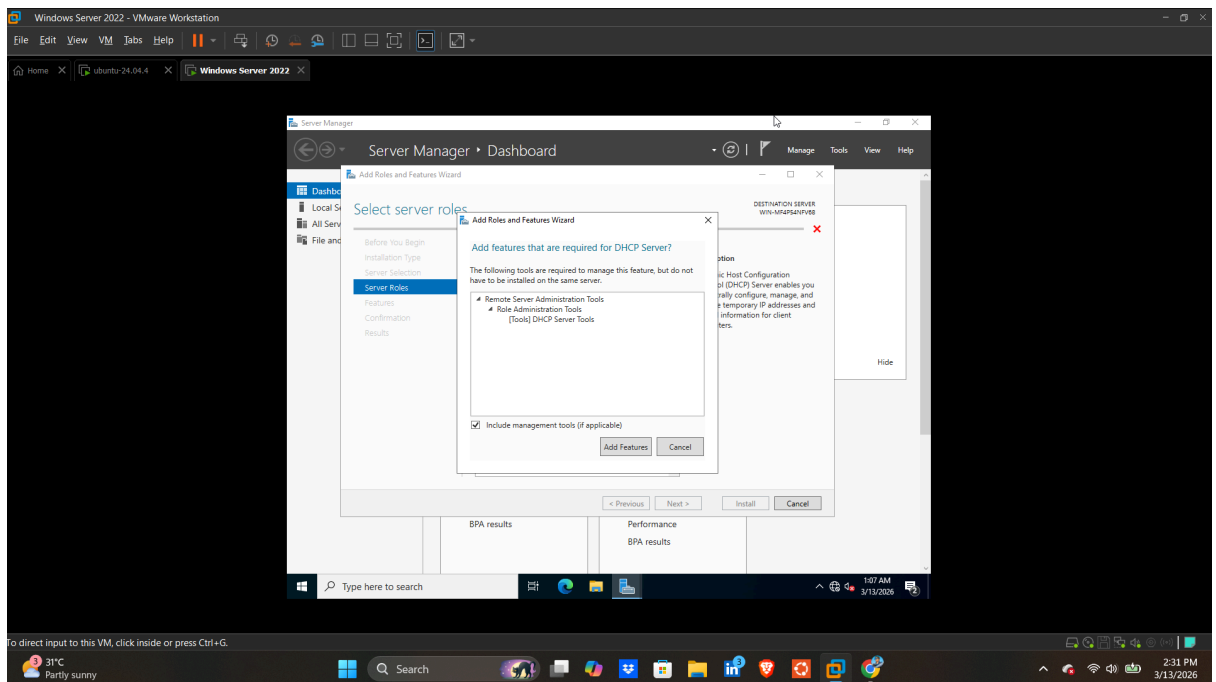
SERVER01
192.168.1.1

Select the server and click **Next**.

5 Selecting the DHCP Server Role

A list of **Server Roles** will appear.

1. Check **DHCP Server**



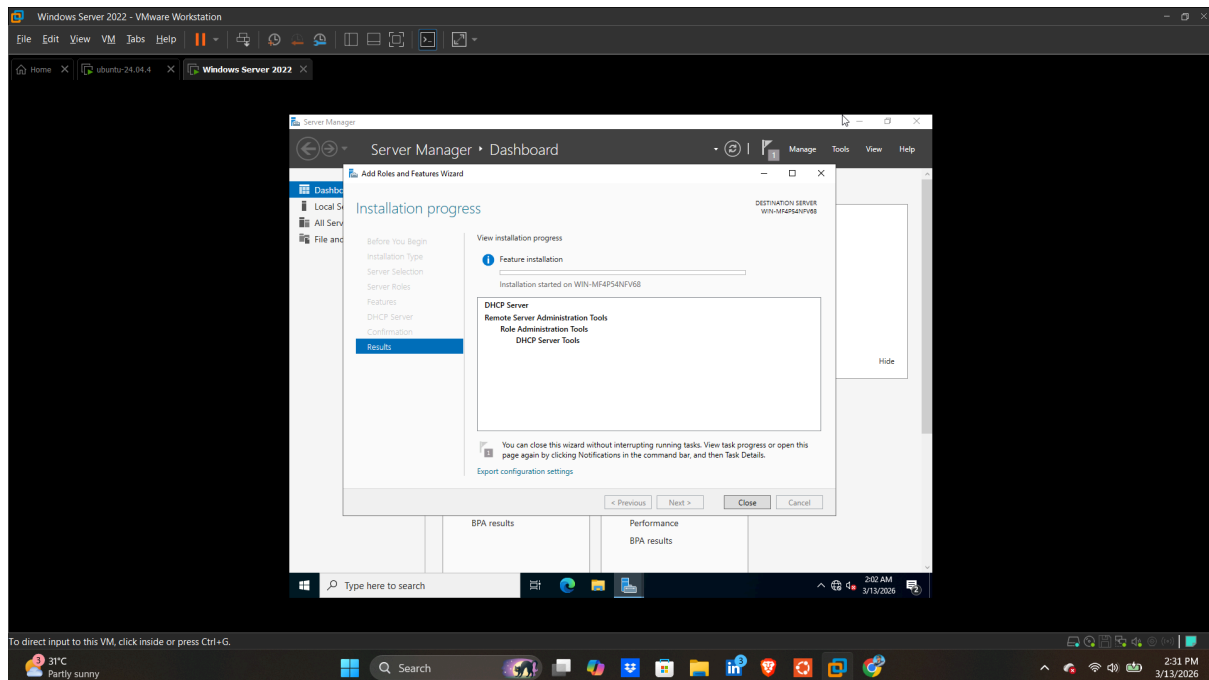
When you select DHCP Server, a popup will appear:

“Add features that are required for DHCP Server?”

2. Click **Add Features**

Then click:

Next → Next → Install



6 Completing the Installation

The installation process may take **1–2 minutes**.

After the installation finishes, a message will appear:

Installation succeeded

Sometimes an option called **DHCP Post-Install Configuration** appears.
If it does:

1. Click **Complete DHCP Configuration**
2. Follow the prompts and click **Finish**

7 Opening the DHCP Management Console

To manage DHCP:

1. Go to **Server Manager**
2. Click **Tools**
3. Select **DHCP**

From the DHCP console you can:

- Create a **DHCP Scope**
- Define the **IP address range**
- Configure **Default Gateway and DNS**
- Automatically assign IP addresses to **client machines**

Step 4: Creating a DHCP Scope (Easy Explanation)

After installing the DHCP server, the next important step is to **create a DHCP Scope**.

A **scope** is a range of IP addresses that the DHCP server can automatically assign to client devices on the network.

Example:

If the scope is **192.168.1.10 – 192.168.1.100**, the DHCP server will assign IP addresses only within this range.

1 Opening the DHCP Console

In **Server Manager**:

Server Manager → **Tools** → **DHCP**

Once the DHCP console opens, you will see a structure on the left side similar to this:

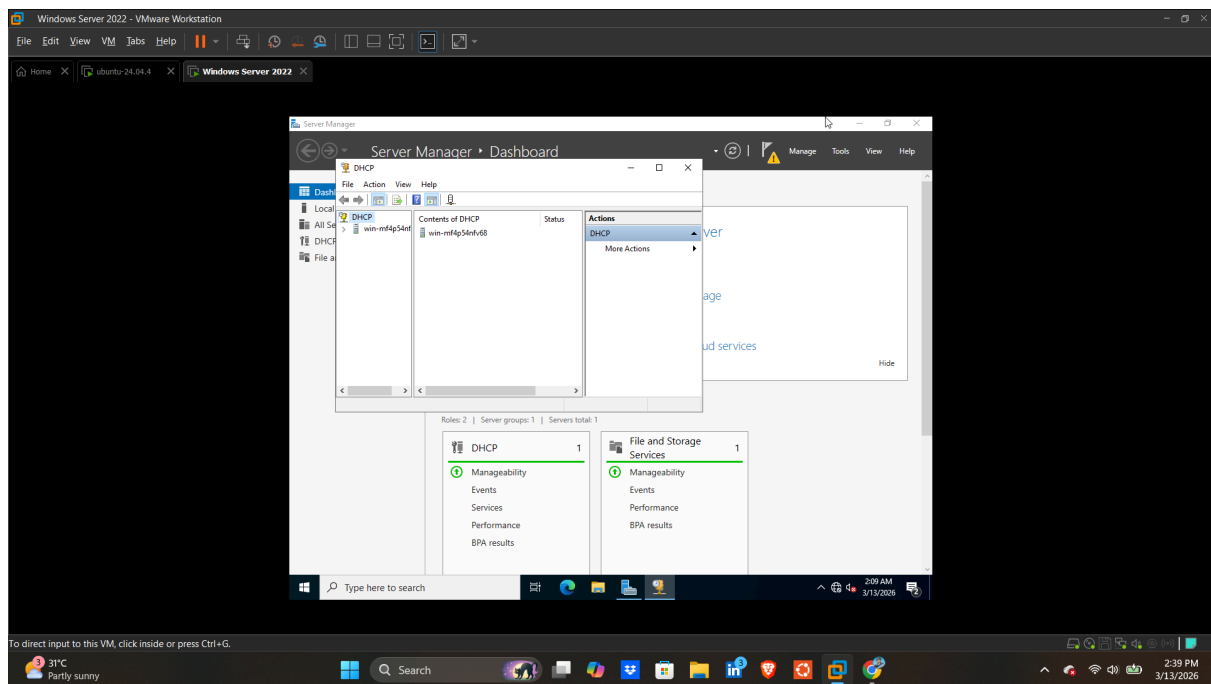
DHCP

- └ Server Name

- └ IPv4

- └ IPv6

Right-click on **IPv4**.

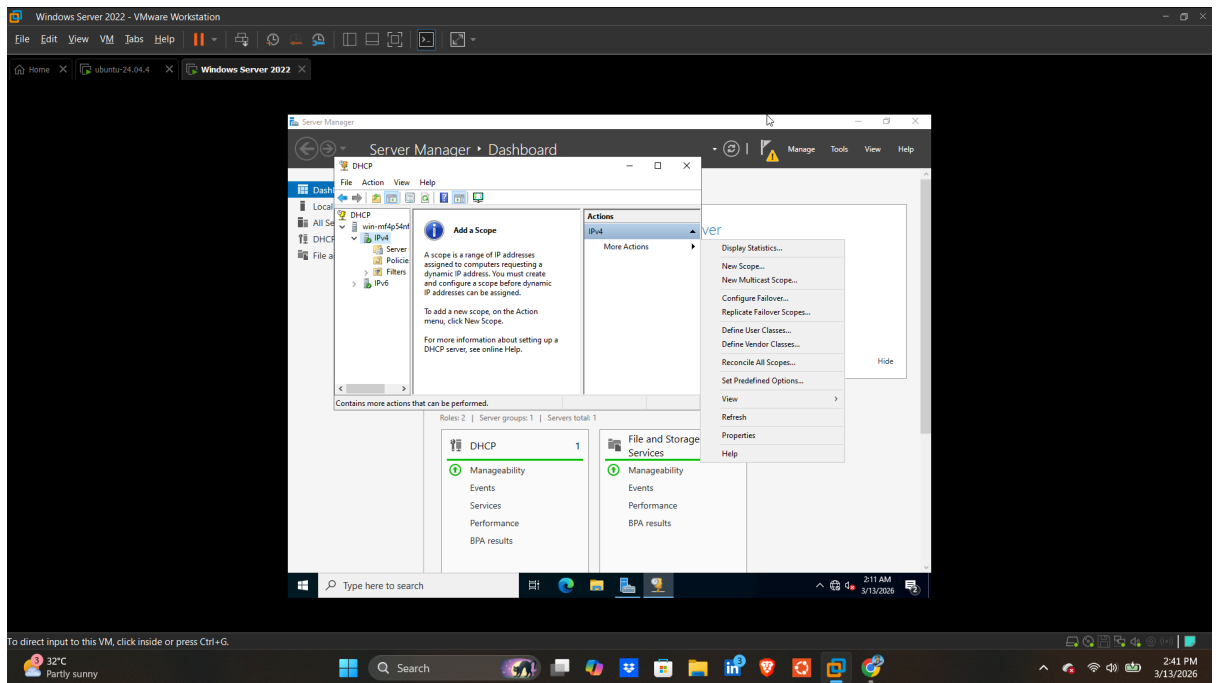


2 Creating a New Scope

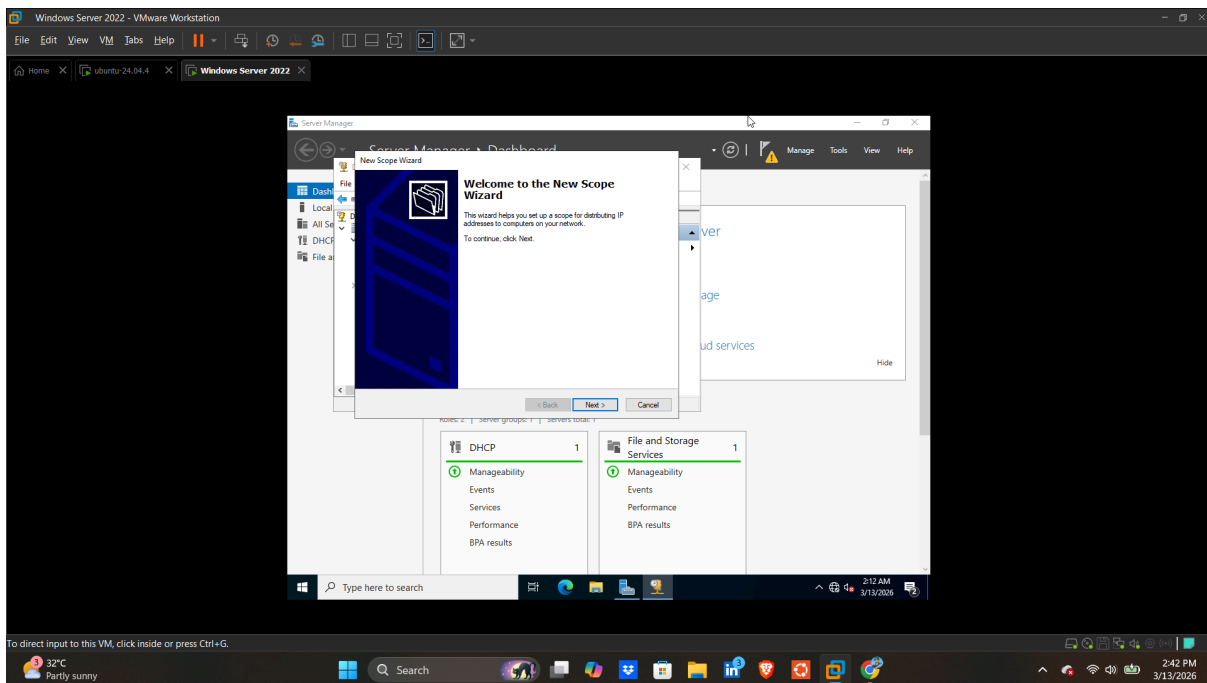
Follow these steps:

1. Right-click on **IPv4**

2. Select New Scope



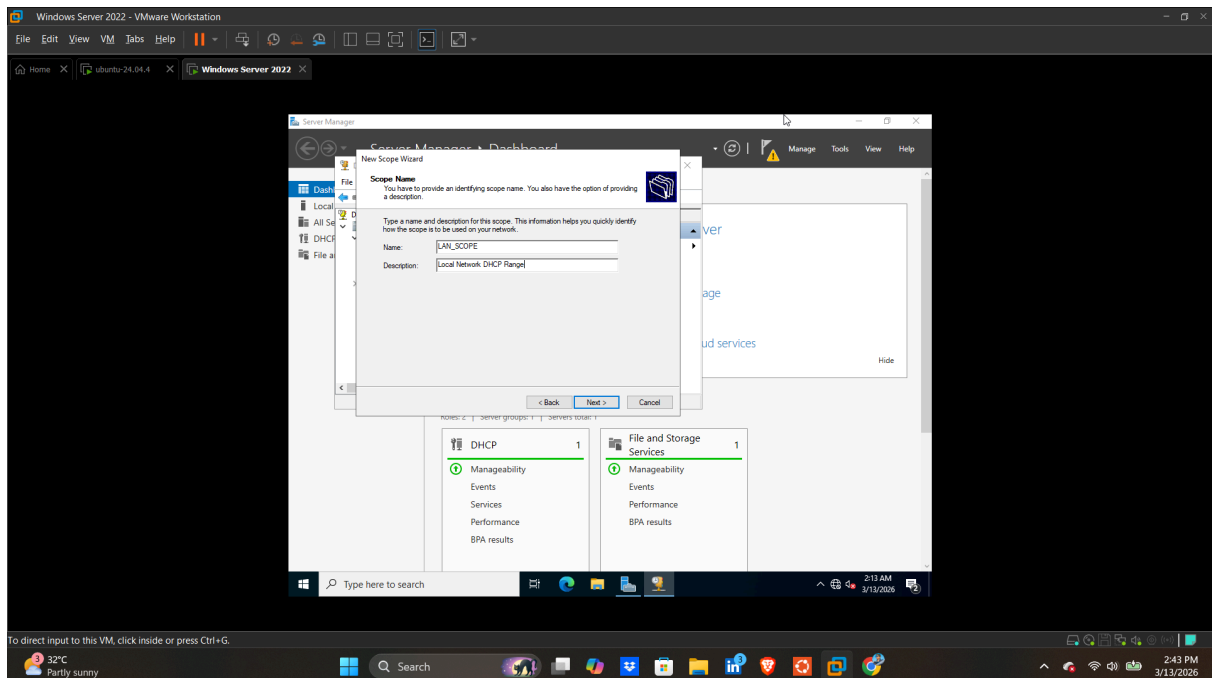
The New Scope Wizard will open.



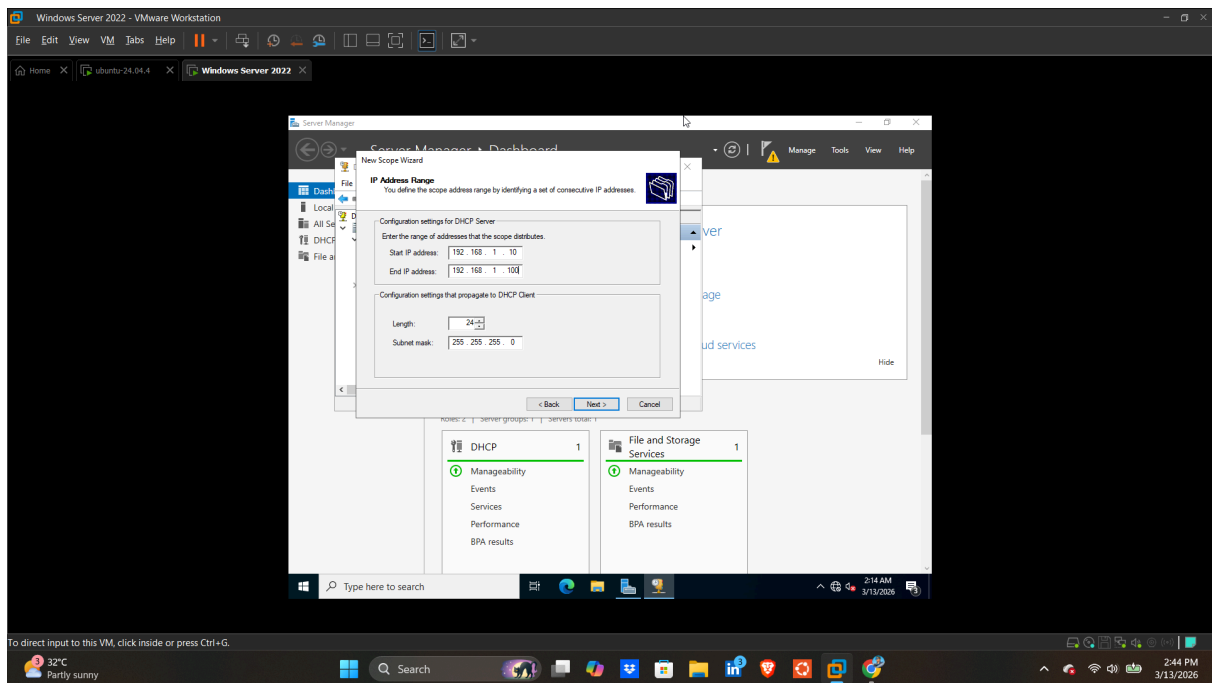
3 Naming the Scope

In the wizard:

- **Scope Name: LAN_SCOPE**



- **Description: Local Network DHCP Range**



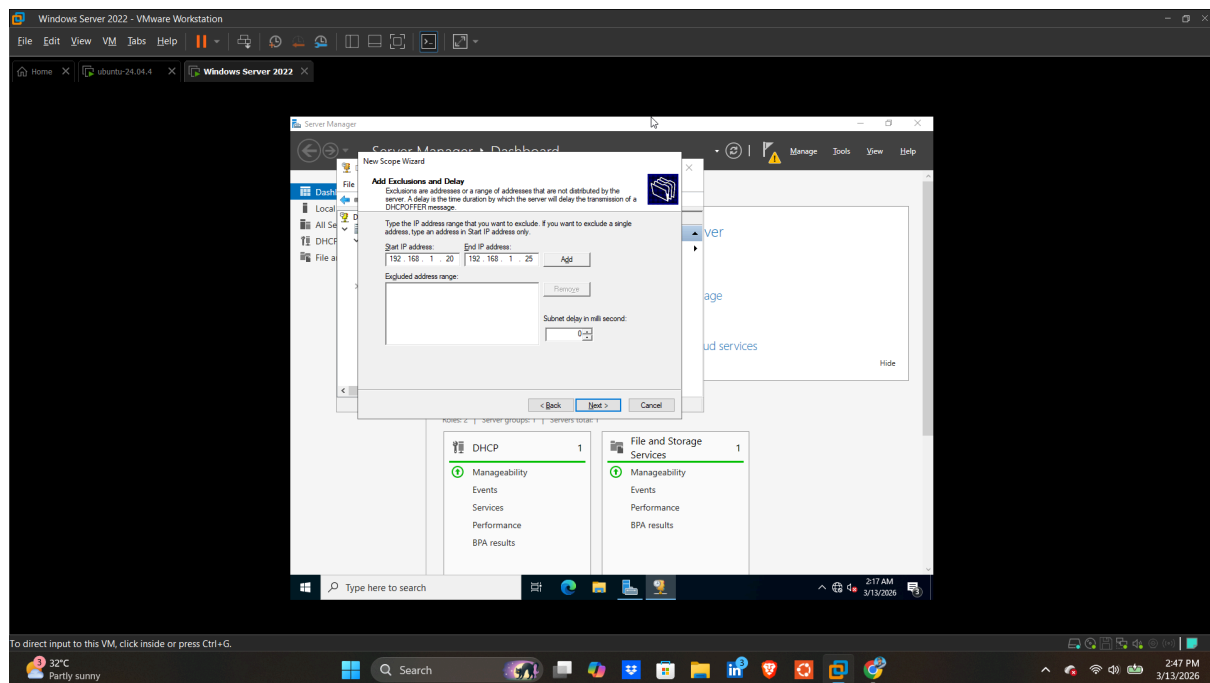
This information is used only for **identification and management purposes**.

5 Exclusion Range (Optional)

In this step, you can exclude specific IP addresses from being assigned by DHCP.

Example:

192.168.1.20 – 192.168.1.25



These addresses will **never be assigned automatically** by the **DHCP server**.

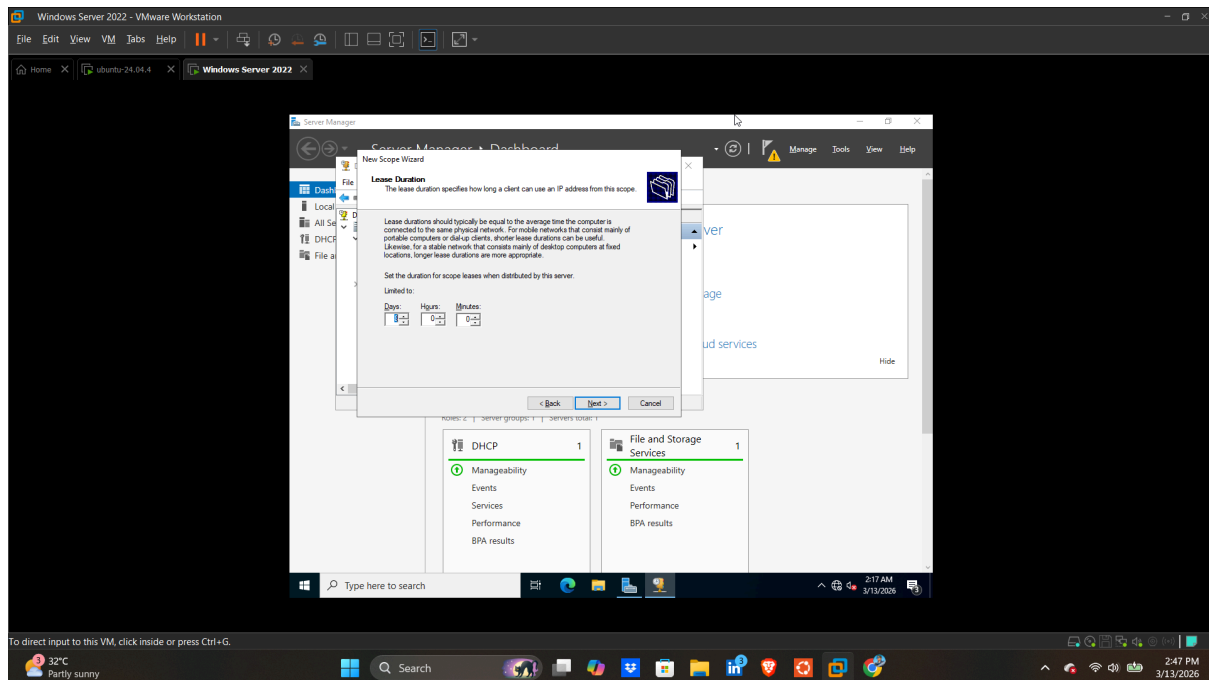
For most lab or project environments, this step can be **skipped**.

6 Lease Duration

The **lease duration** determines how long a client device can use the assigned IP address.

Default setting:

8 Days



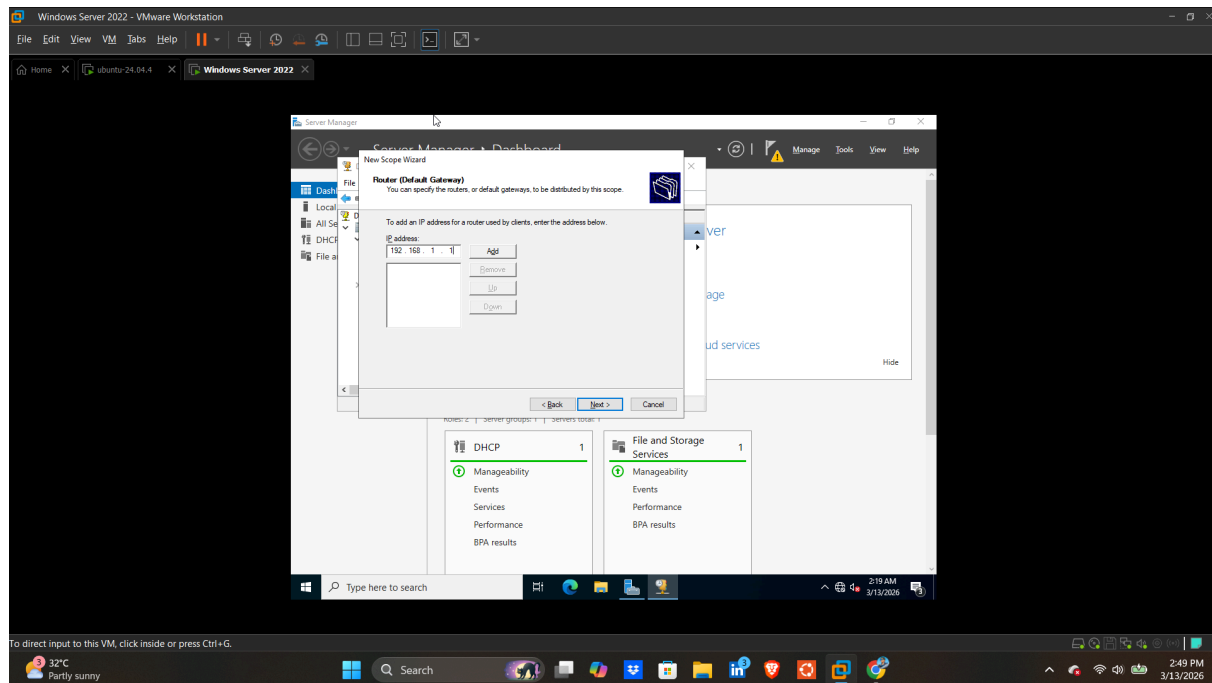
This means the client device will keep the same IP address for **eight days** before it must renew the lease.

7 Configuring the Default Gateway

Next, set the **Default Gateway (Router address)**.

Example:

192.168.1.1



This is usually the **router** or the **main network gateway**.

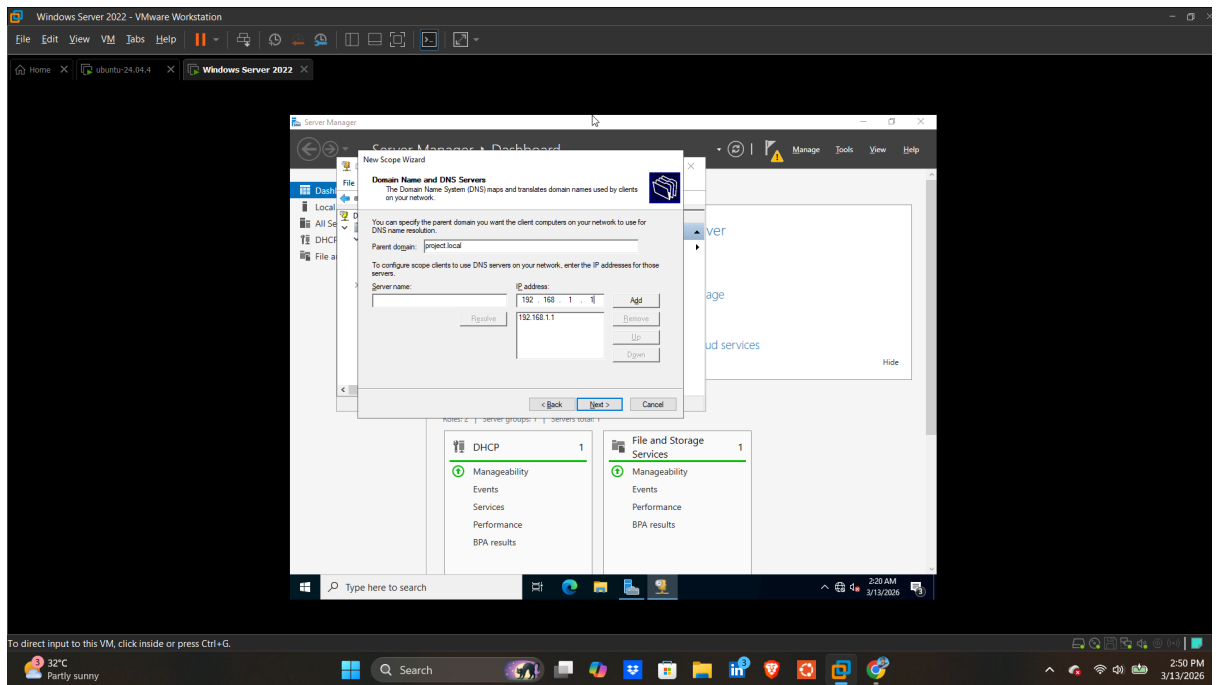
8 Configuring the DNS Server

Here you specify the DNS server that clients will use.

Example:

- **Parent Domain:** project.local

- **DNS Server: 192.168.1.1**



This means that clients will send **DNS queries** to the server at **192.168.1.1**.

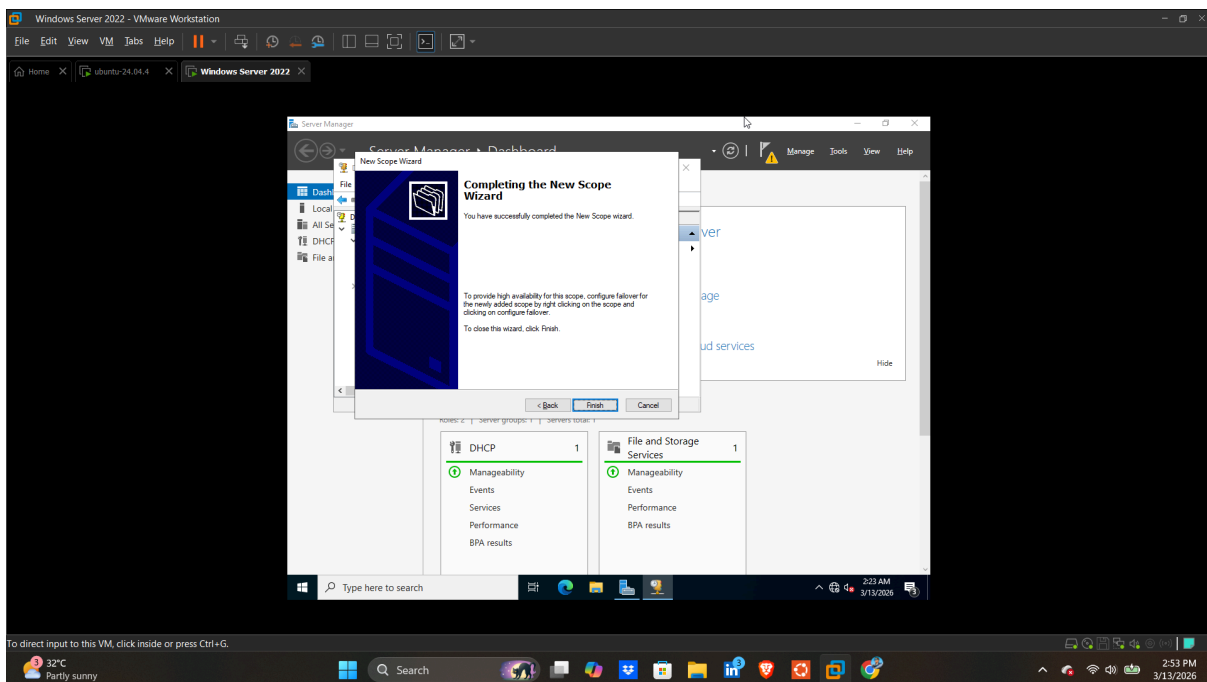
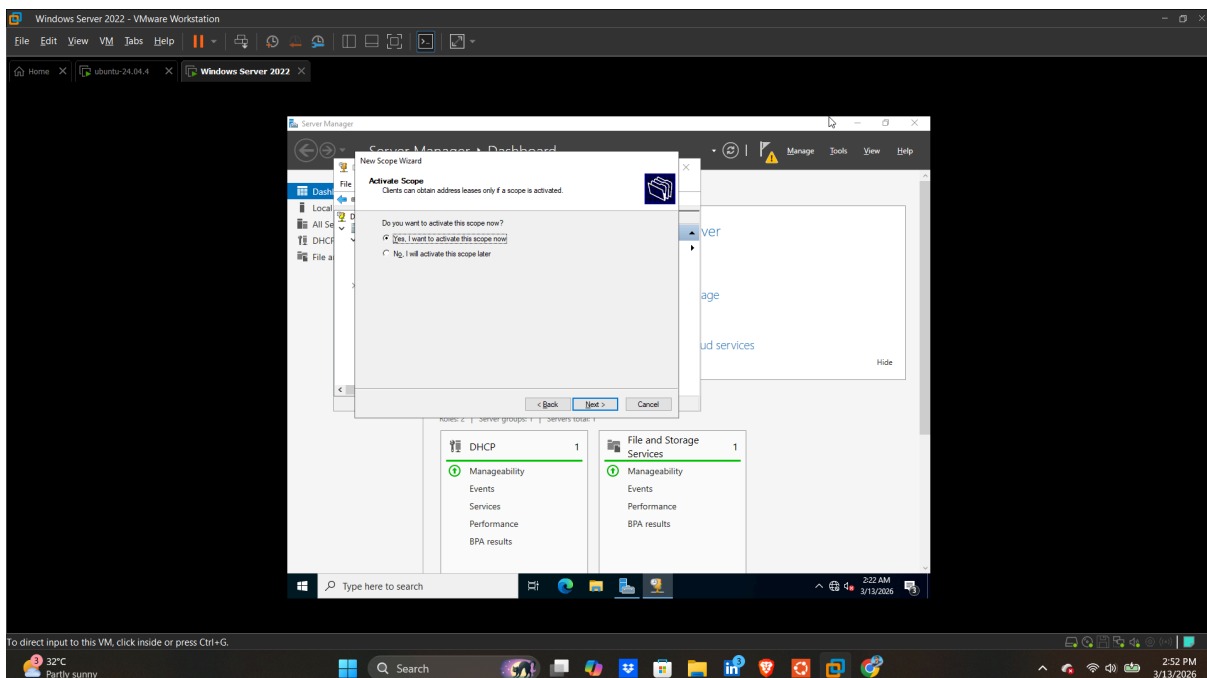
9 Activating the Scope

In the final step, the wizard will ask:

“Do you want to activate this scope now?”

Select:

✓ Yes



Then click **Finish**.

10 After the Scope is Created

The DHCP console will now show the scope structure:

DHCP

└ Server

└ IPv4

└ LAN_SCOPE

└ Address Pool

└ Address Leases

When a client device (for example, an Ubuntu machine) sends a DHCP request, the server will automatically assign an IP address.

Example configuration assigned to a client:

- **Ubuntu Client IP:** 192.168.1.10
- **DNS Server:** 192.168.1.1
- **Default Gateway:** 192.168.1.1

In Simple Terms

A **DHCP Scope** is an **IP address pool**.

This means the DHCP server will automatically assign IP addresses to client devices **within the defined range (192.168.1.10 – 192.168.1.100)**.

Step 5: Installing the DNS Server (Windows Server 2022)

The goal of this step is to **install the DNS Server and create a domain zone** so that computers on the network can connect using **hostnames instead of IP addresses**.

Example:

 `ping 192.168.1.1`

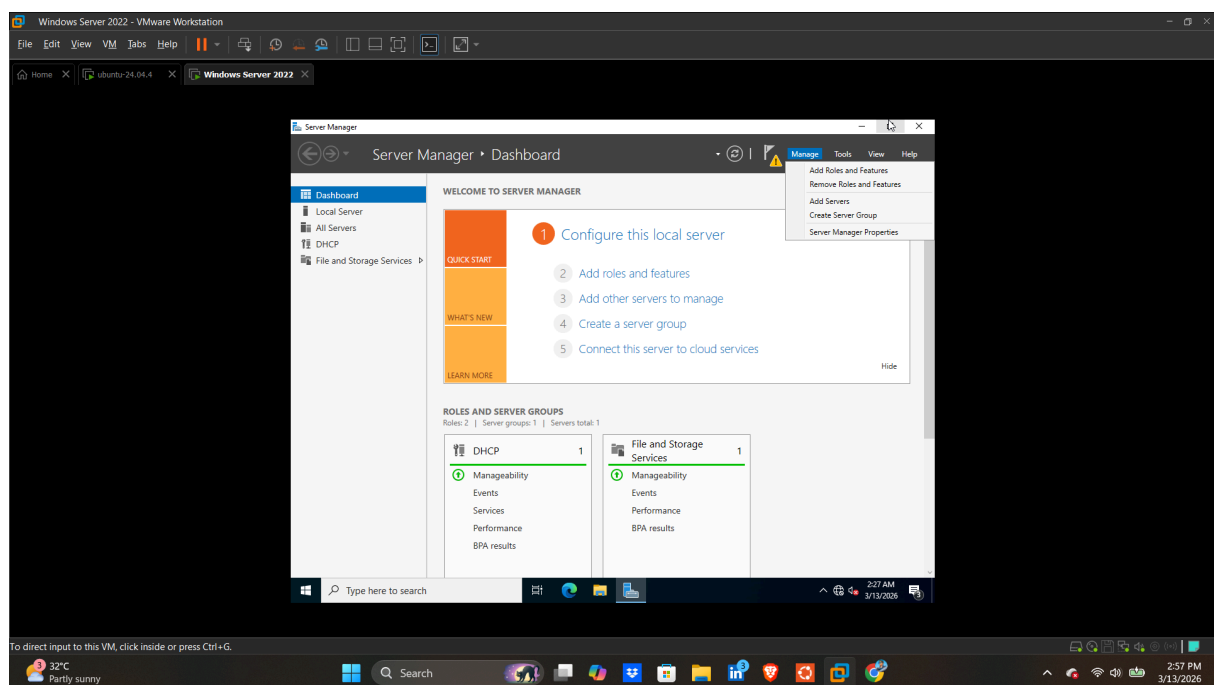
 `ping server.project.local`

The function of **DNS (Domain Name System)** is to convert a **domain name** into an **IP address**.

1 Installing the DNS Server

Follow these steps:

1. Open **Server Manager**
2. Click **Manage** in the top menu
3. Select **Add Roles and Features**



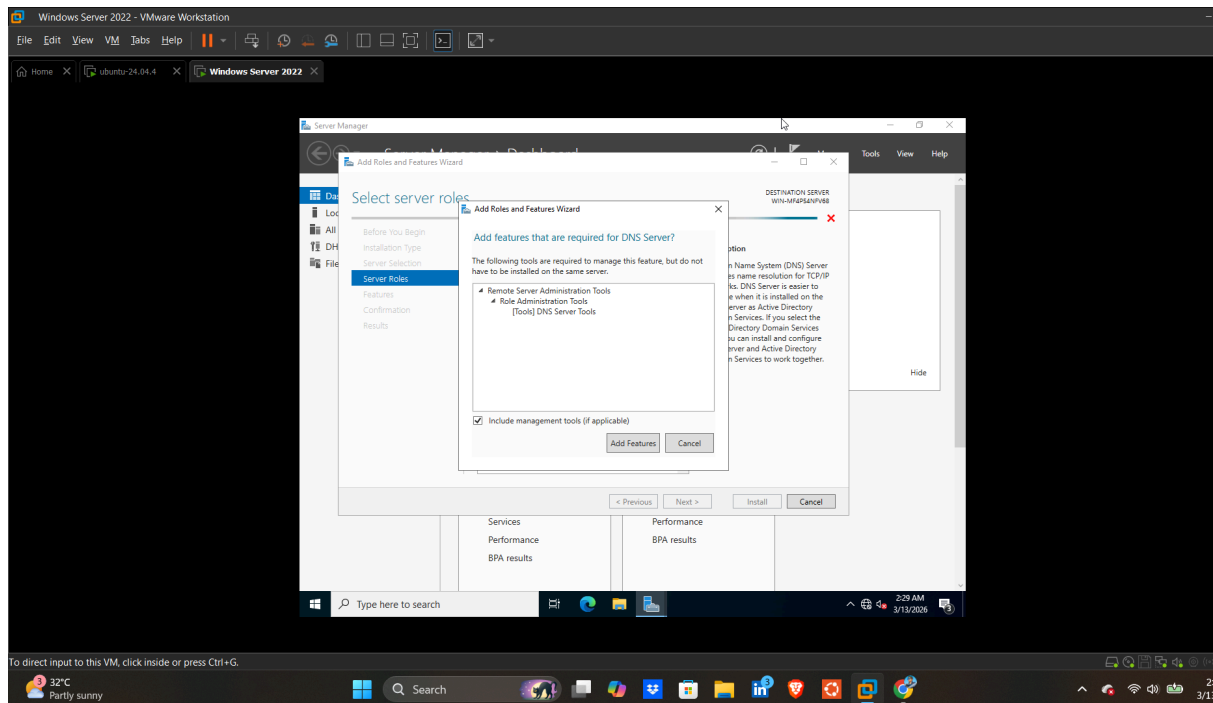
The installation wizard will open.

Next steps:

1. Select **Role-based or feature-based installation**
2. Click **Next** → **Next**

A list of **Server Roles** will appear.

✓ Select DNS Server

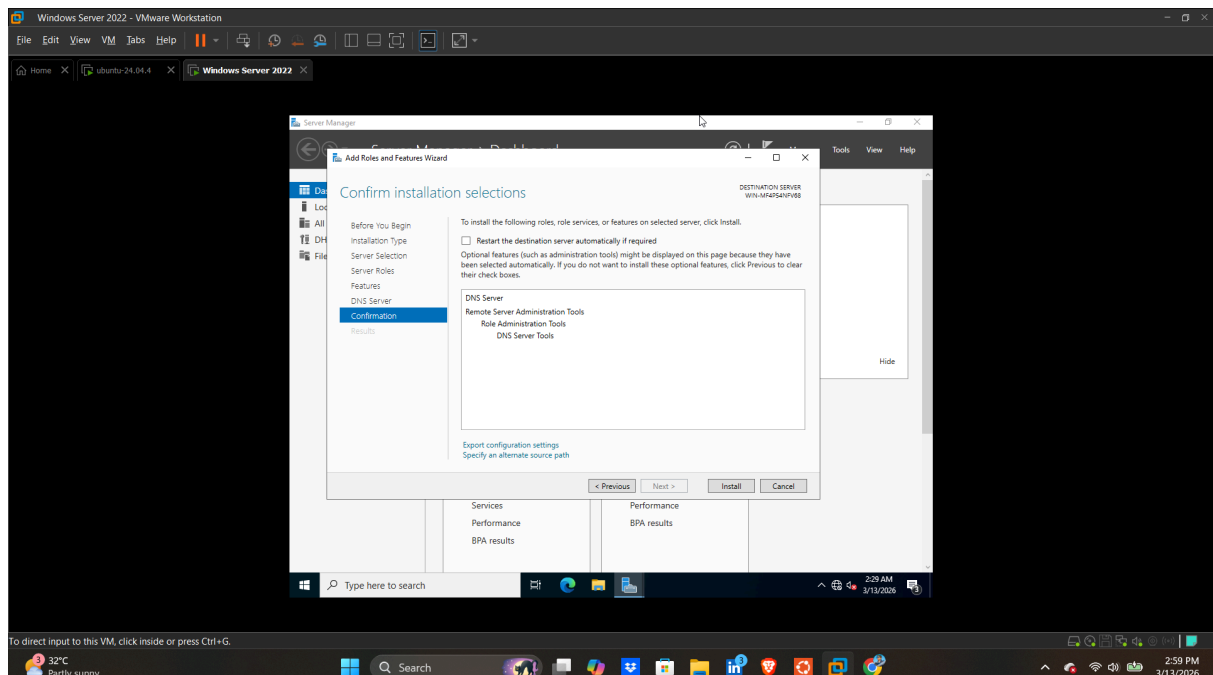


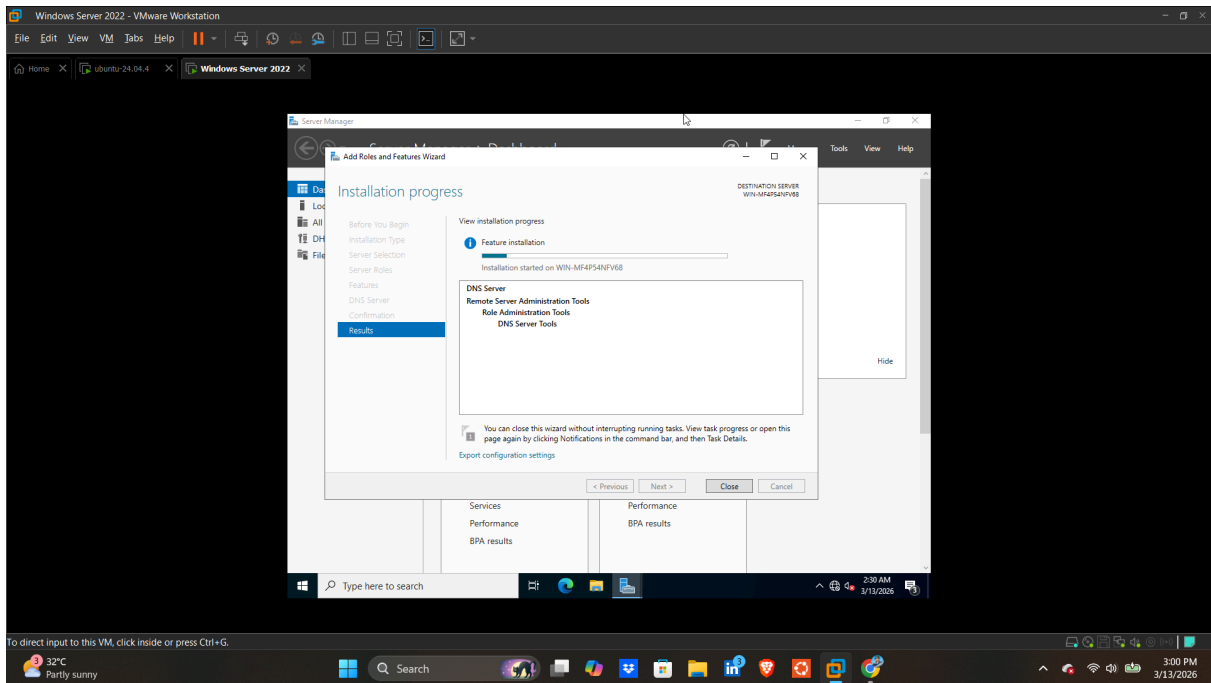
A popup will appear:

“Add features that are required for DNS Server?”

👉 Click **Add Features**

👉 Then click **Next → Install**



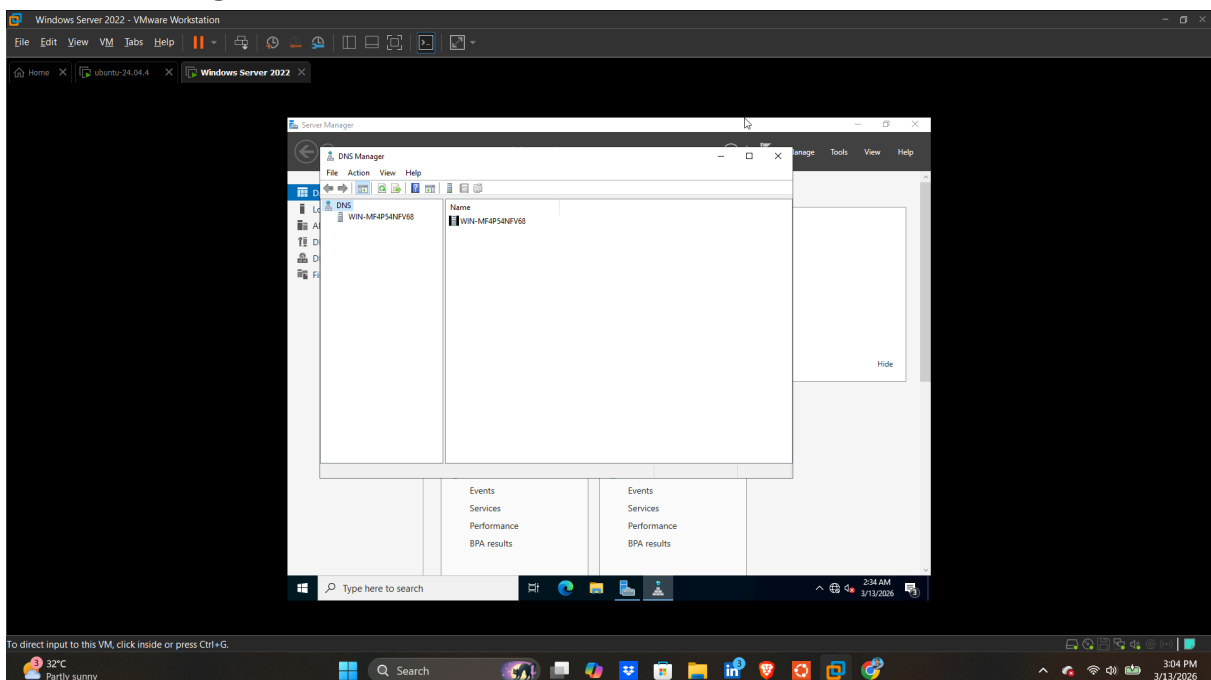


Once the installation is completed, click **Close**.

2 Opening DNS Manager

To manage DNS:

Server Manager → Tools → DNS



The **DNS Manager** will open.

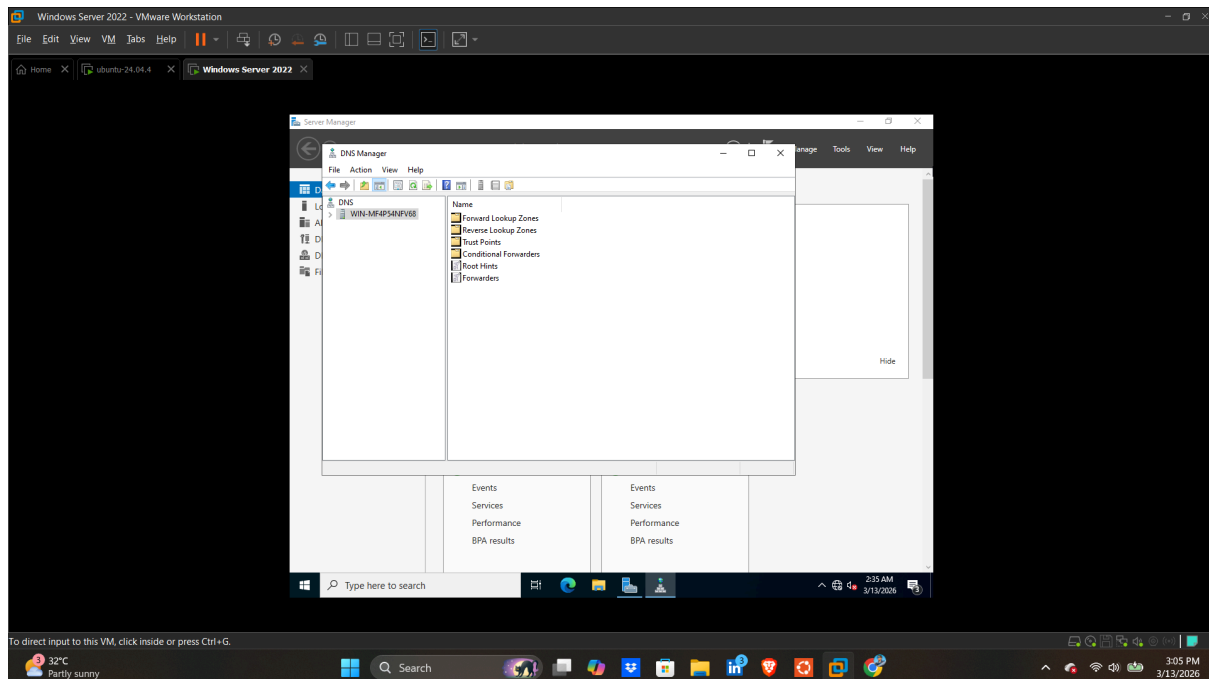
The structure will appear similar to this:

DNS

- └ Server Name

 - └ Forward Lookup Zones

 - └ Reverse Lookup Zones



3 Creating a Forward Lookup Zone

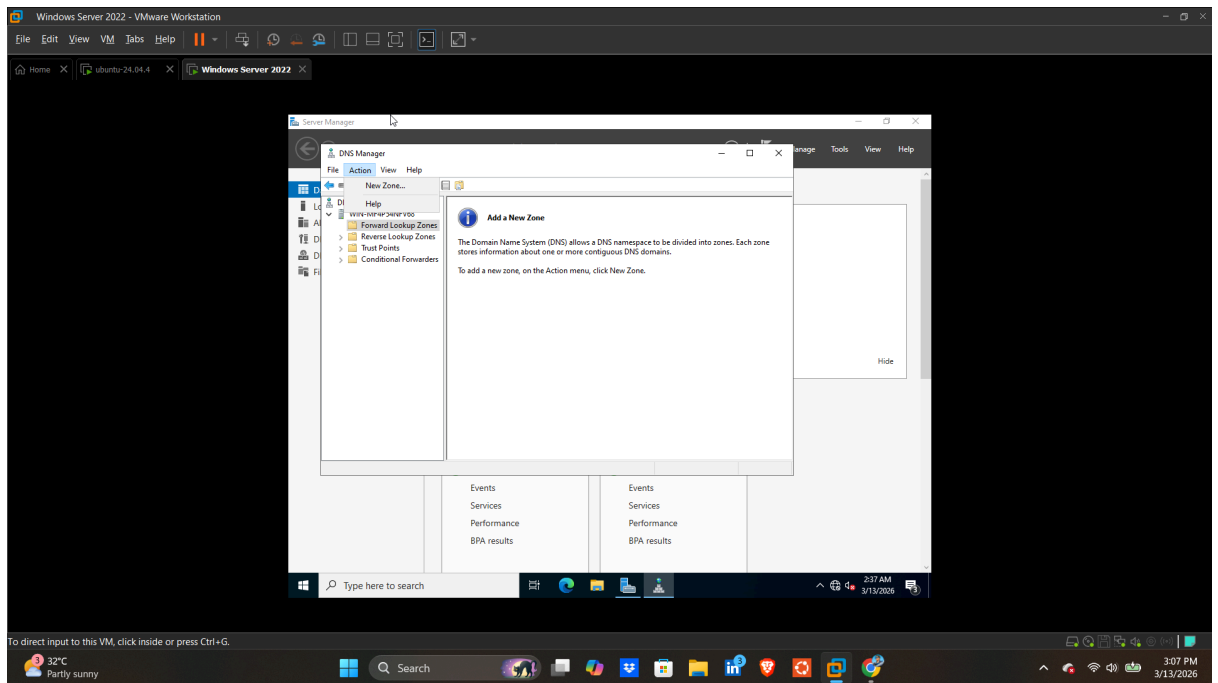
A **Forward Lookup Zone** is used for converting:

Domain Name → **IP Address**

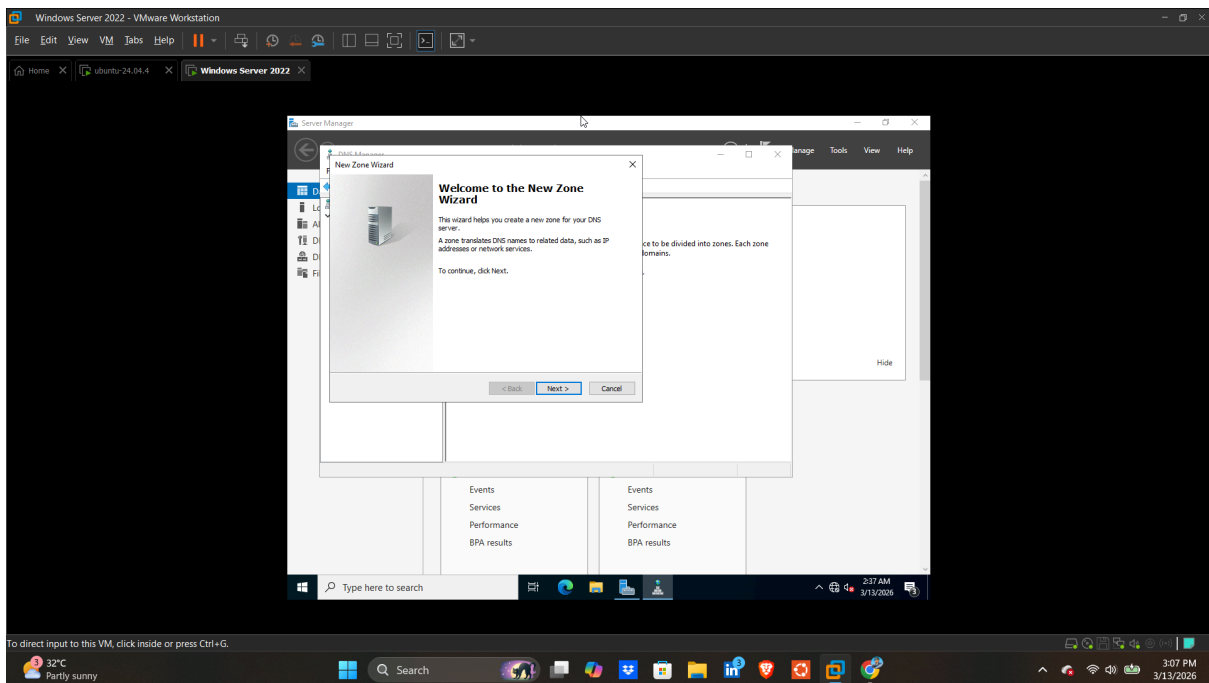
Steps:

1. Right-click **Forward Lookup Zones**

2. Select New Zone

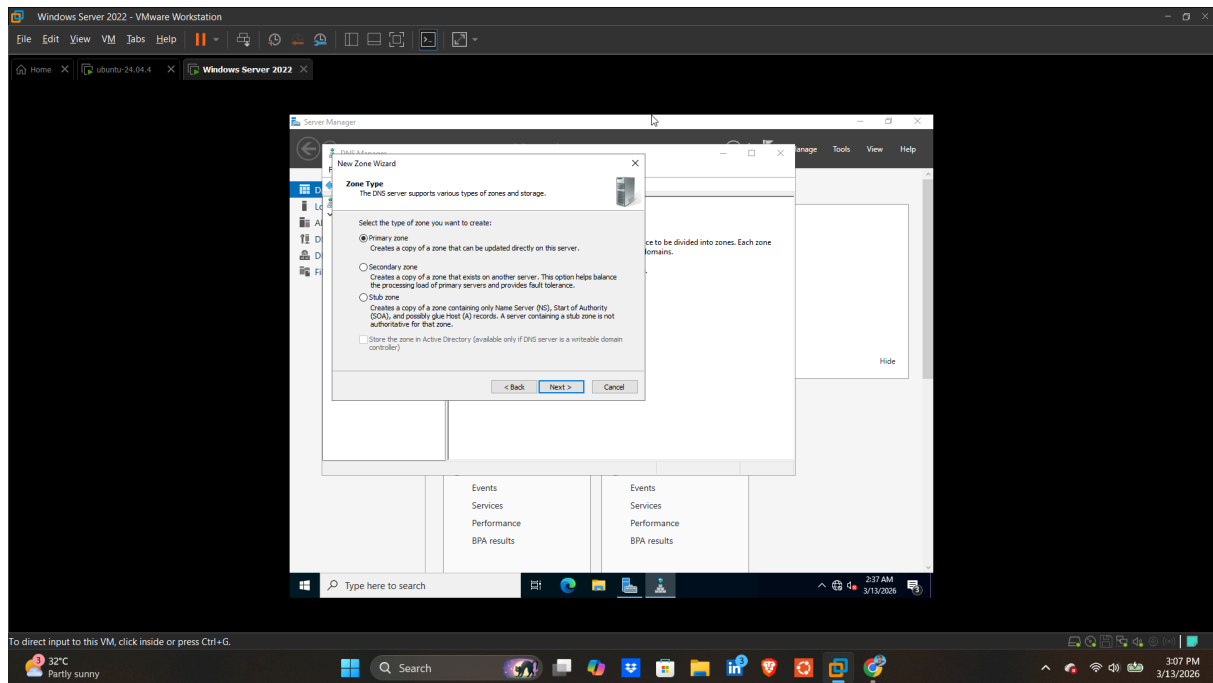


The New Zone Wizard will open.



Select:

✓ Primary Zone



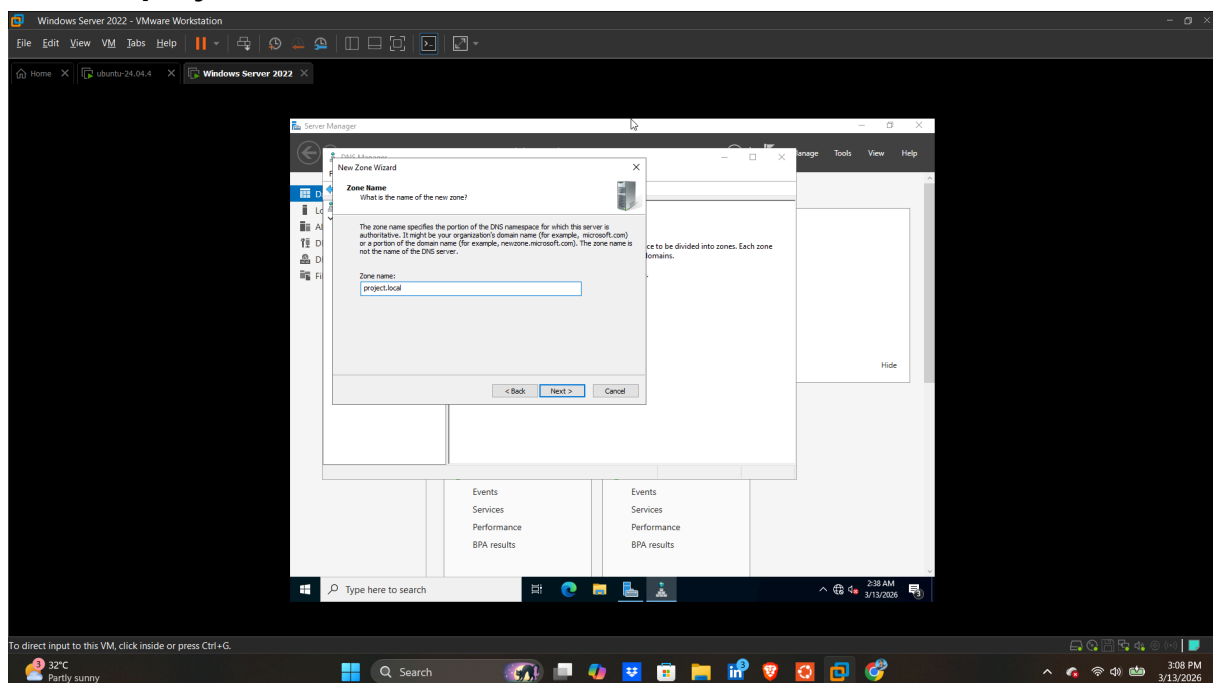
Click **Next**.

Enter the **Zone Name**:

project.local

Click **Next** → **Finish**

Now the **project.local** zone will be created.

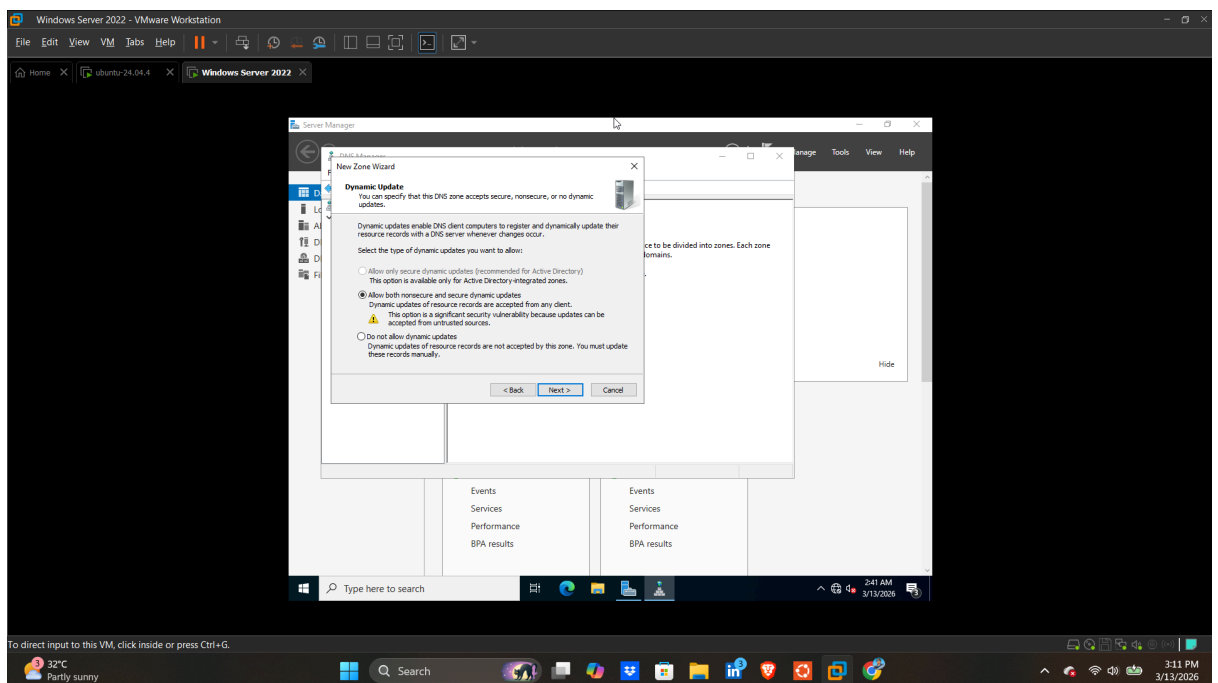
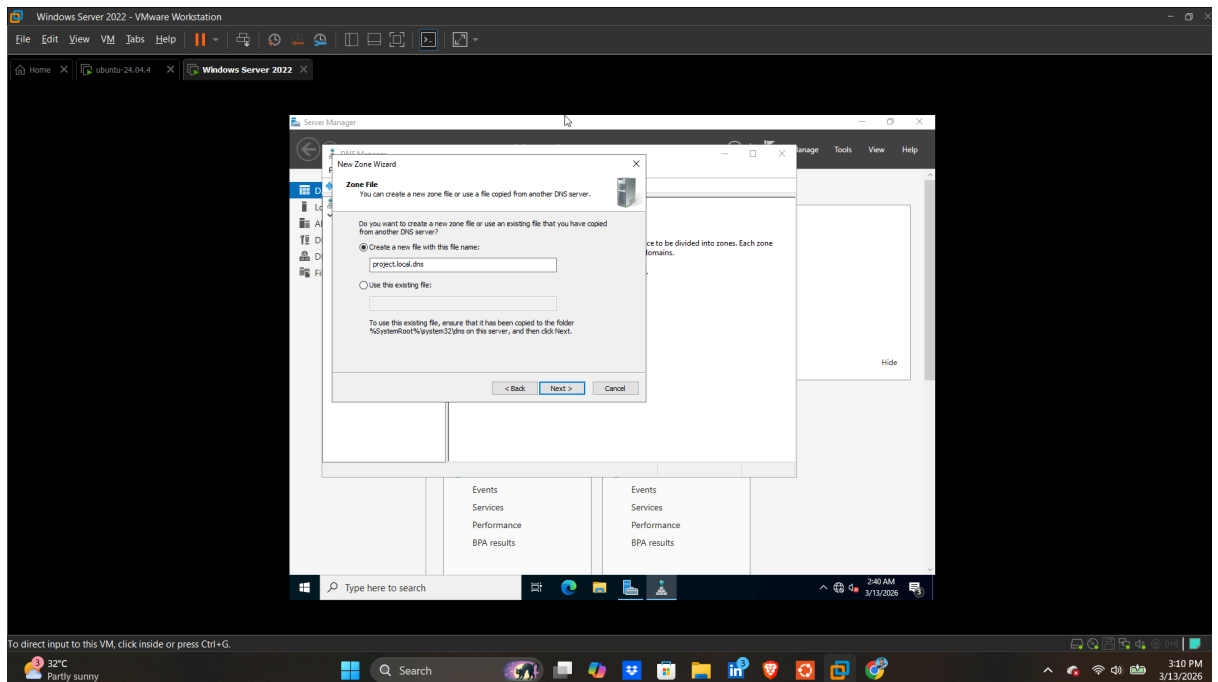


4 Adding a DNS Record

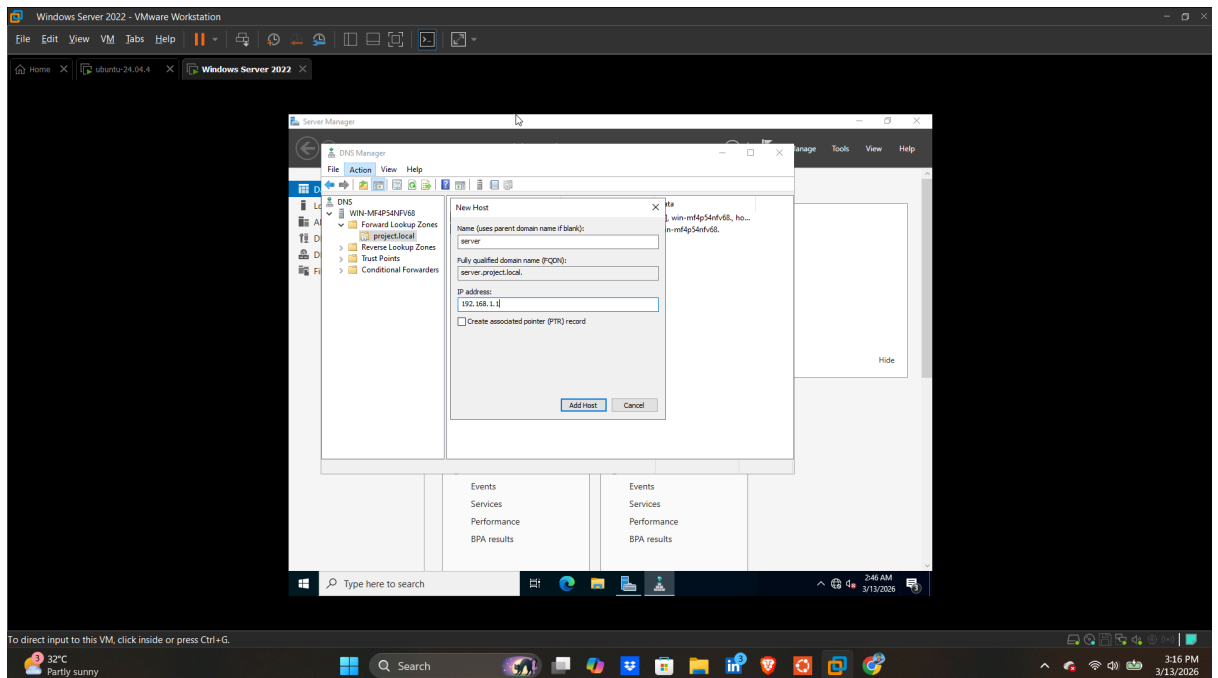
Next, we need to create a **DNS record for the server**.

Steps:

1. Right-click **project.local**



2. Select **New Host (A or AAAA)**



Enter the following information:

Field	Value
Name	server
IP Address	192.168.1.1

Example:

server.project.local → 192.168.1.1

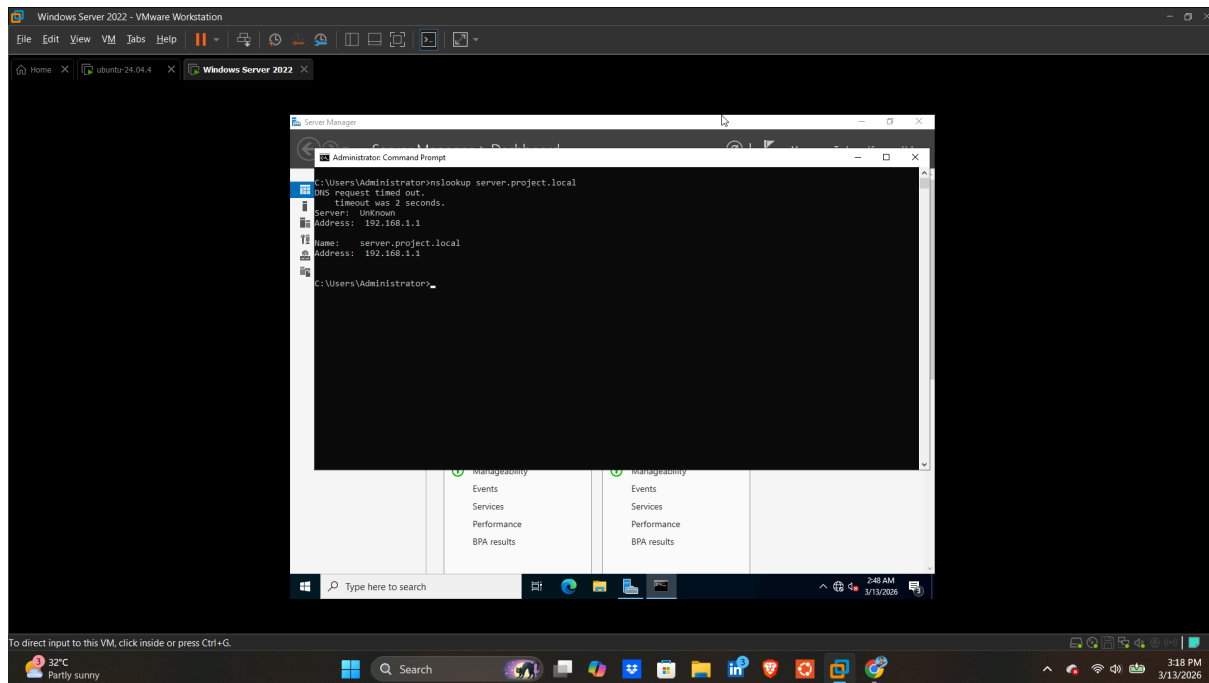
Click **Add Host**.

The DNS record is now created.

5 Testing the DNS Configuration (Important)

Run the following command on the server or client machine:

nslookup server.project.local



Expected Result

Name: server.project.local

Address: 192.168.1.1

This confirms that the **DNS server is functioning correctly.**

Simple Example (Real Meaning)

Domain Name	IP Address
server.project.local	192.168.1.1
google.com	142.250.x.x

The purpose of DNS is to **translate domain names into IP addresses.**

Step 6: Enabling DHCP on the Ubuntu Client

The goal of this step is to configure the **Ubuntu client to automatically obtain an IP address from the Windows Server DHCP server.**

In this setup:

- **Windows Server** acts as the **DHCP server (192.168.1.1)**
 - **Ubuntu** acts as the **client machine**
-

1 Checking the Current IP Address on Ubuntu

Open the **Ubuntu virtual machine** and launch the **Terminal**.

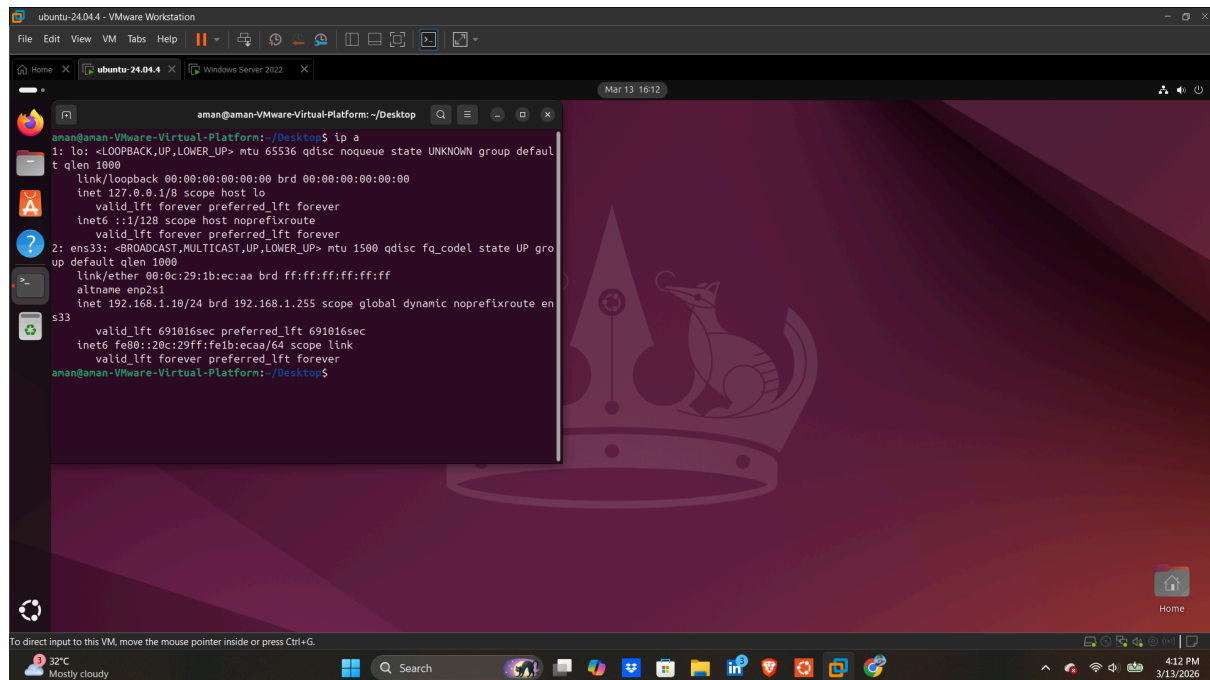
Run the following command:

```
ip a
```

This command displays the **network interfaces and their assigned IP addresses**.

Example output:

eth0: inet 192.168.1.10/24



If there is **no IP address** or the address appears as **169.254.x.x**, it means the system did not receive an IP address from the DHCP server.

2 Opening Network Settings

In Ubuntu:

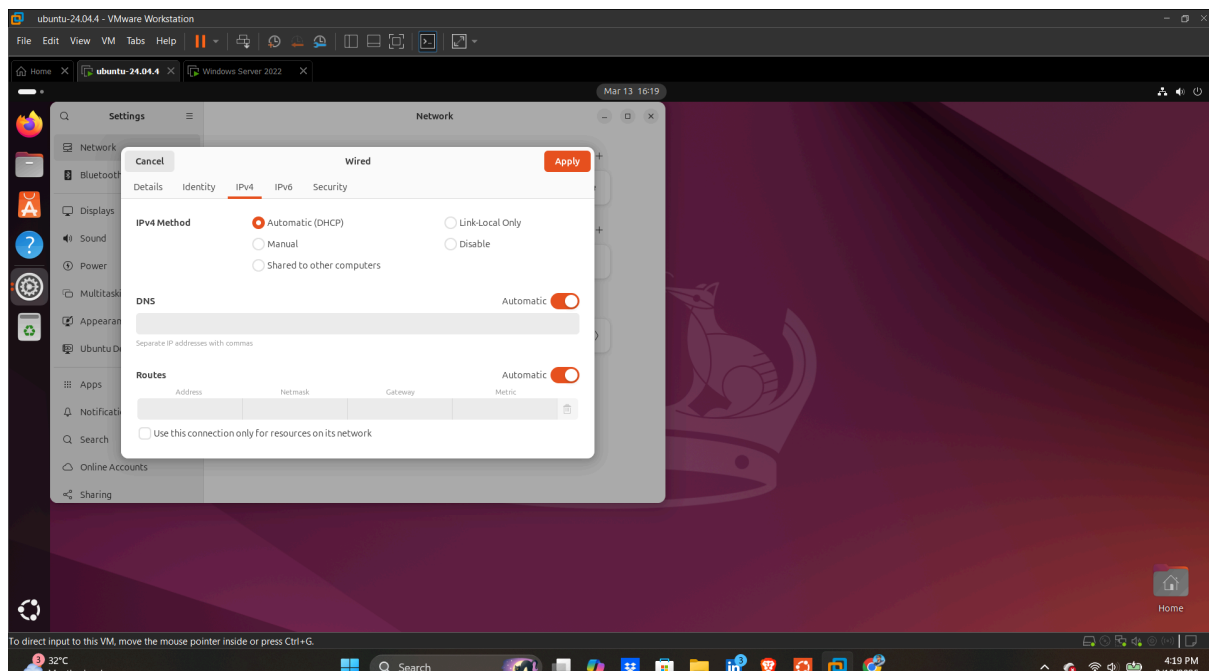
1. Click the **network icon** in the **top-right corner** of the screen
2. Open **Wired Connected / Settings**
3. Navigate to **Network Settings**

Then select:

Wired Connection → Settings (⚙ icon)

3 Setting the IPv4 Method

Open the **IPv4** tab.



You will see an option called **Method**.

Select:

Automatic (DHCP)

This configuration allows Ubuntu to **automatically request an IP address from the DHCP server**.

After selecting the option, click **Apply / Save**.

4 Renewing the DHCP Lease

Open the terminal and run the following command:

```
sudo dhclient
```

This command sends a request to the **DHCP server** for a new IP address.

This process follows the **DORA process**:

- **Discover**

- Offer
 - Request
 - Acknowledge
-

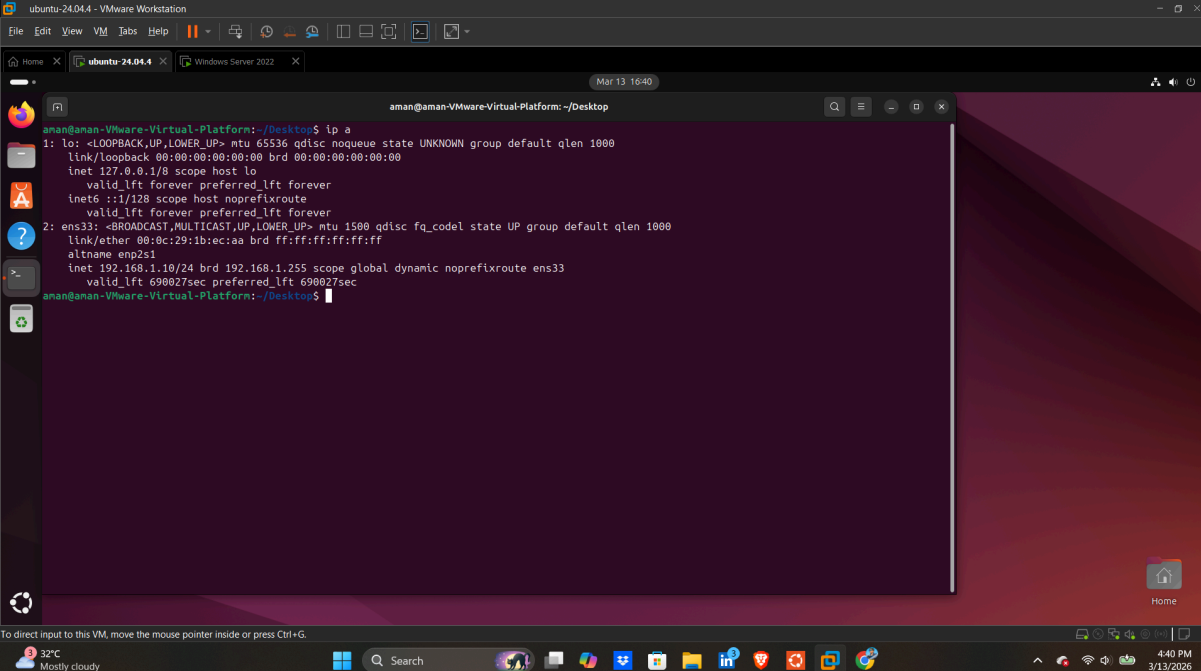
5 Verifying the Assigned IP Address

Run the following command again:

ip a

Expected output:

eth0: inet 192.168.1.10/24



```
aman@aman-VMware-Virtual-Platform: ~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1b:ec:aa brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic noprefixroute ens33
        valid_lft 690027sec preferred_lft 690027sec
aman@aman-VMware-Virtual-Platform: ~/Desktop$
```

This indicates that the **Ubuntu client** has successfully received an IP address from the DHCP server.

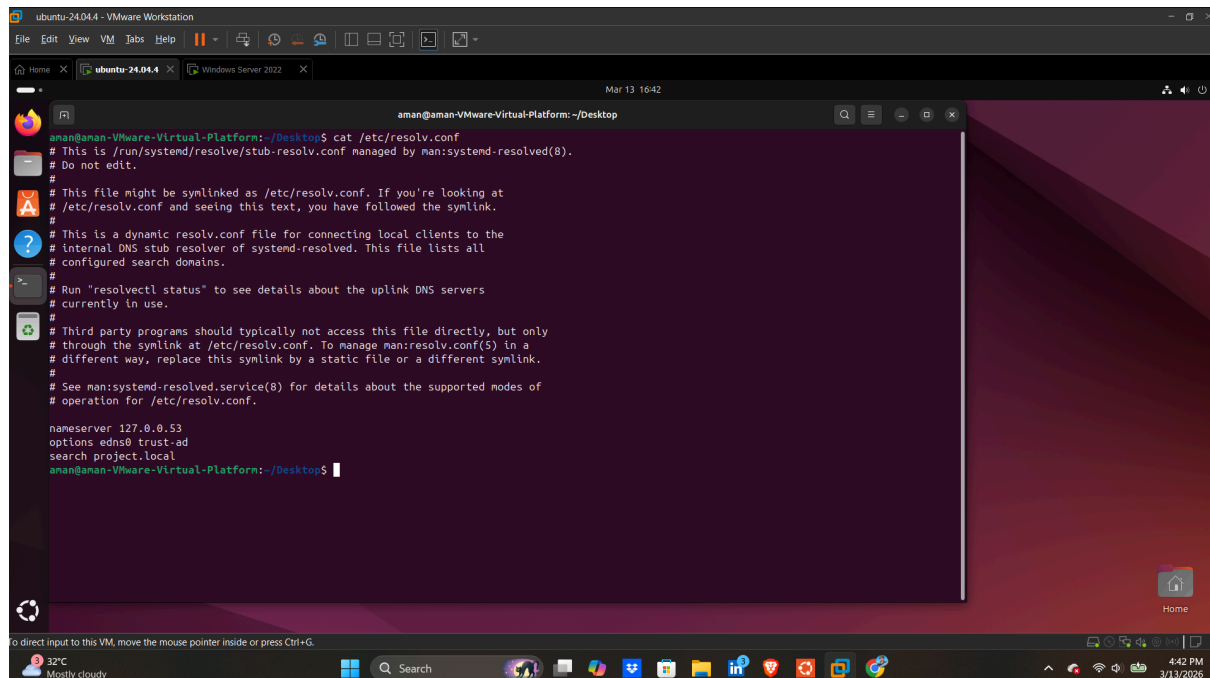
6 Checking the DNS Configuration

To verify the DNS configuration, run:

```
cat /etc/resolv.conf
```

Expected output:

```
nameserver 192.168.1.1
```



```
aman@aman-VMware-Virtual-Platform: ~/Desktop
aman@aman-VMware-Virtual-Platform:~/Desktop$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
nameserver 127.0.0.53
options edns0 trust-ad
search project.local
aman@aman-VMware-Virtual-Platform:~/Desktop$
```

This confirms that **Ubuntu is using the Windows Server as its DNS server.**

The Ubuntu client should receive the following configuration from the DHCP

server:

```
aman@aman-Virtual-Platform: ~/Desktop
aman@aman-Virtual-Platform:~/Desktop$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
search project.local
aman@aman-Virtual-Platform:~/Desktop$ nslookup server.project.local
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   server.project.local
Address: 192.168.1.1

aman@aman-Virtual-Platform:~/Desktop$
```

Setting	Value
IP Address	192.168.1.1 0
Subnet Mask	255.255.25 5.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.1

Four Important command to prove the project

ip a

ping 192.168.1.1

cat /etc/resolv.conf

nslookup server.project.local

Step 7: Network Testing

In this step, three tests are performed to verify that the **network configuration, DHCP service, and DNS functionality** are working correctly.

1 Ping Test (Server Connectivity Test)

Command:

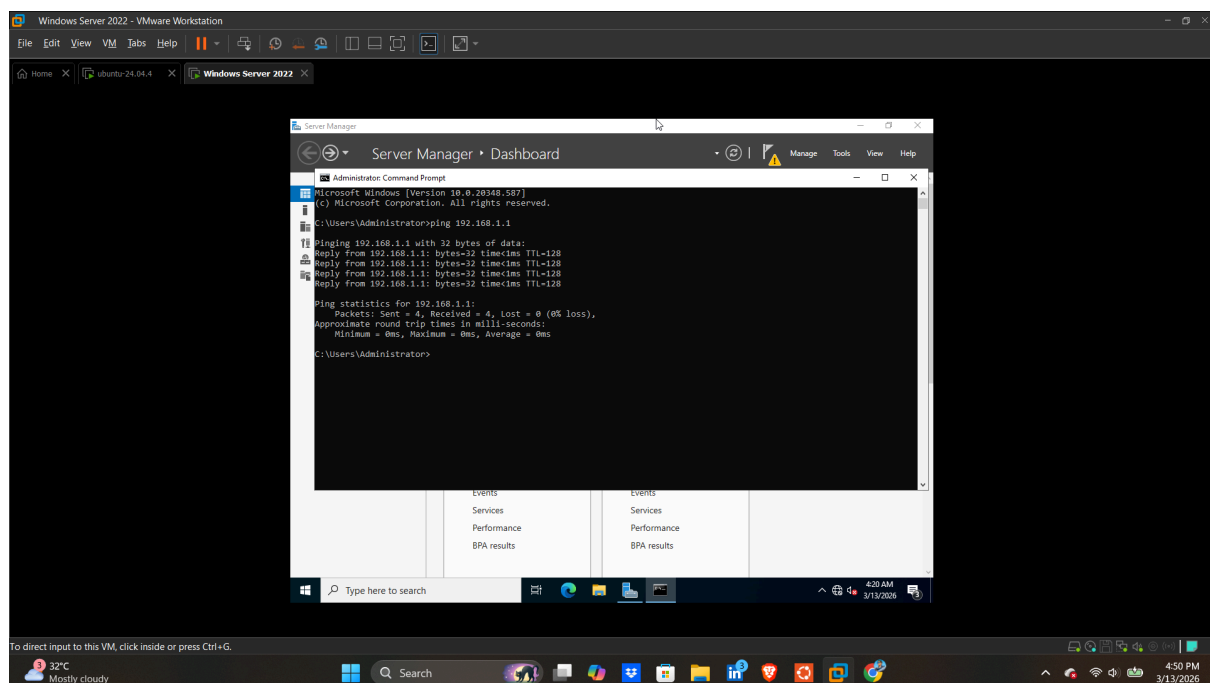
```
ping 192.168.1.1
```

Purpose

This command checks whether the **Ubuntu client can reach the Windows Server** over the network.

Here:

192.168.1.1 → IP address of the Windows Server



Expected Output

64 bytes from 192.168.1.1: icmp_seq=1 ttl=128 time=1 ms

Meaning

✓ The network connection between the **Ubuntu client** and the **Windows Server** is working correctly.

If replies are received, it confirms that the **server is reachable from the client machine**.

To stop the ping process, press:

Ctrl + C

2 DNS Test (Domain Name Resolution Test)

Command:

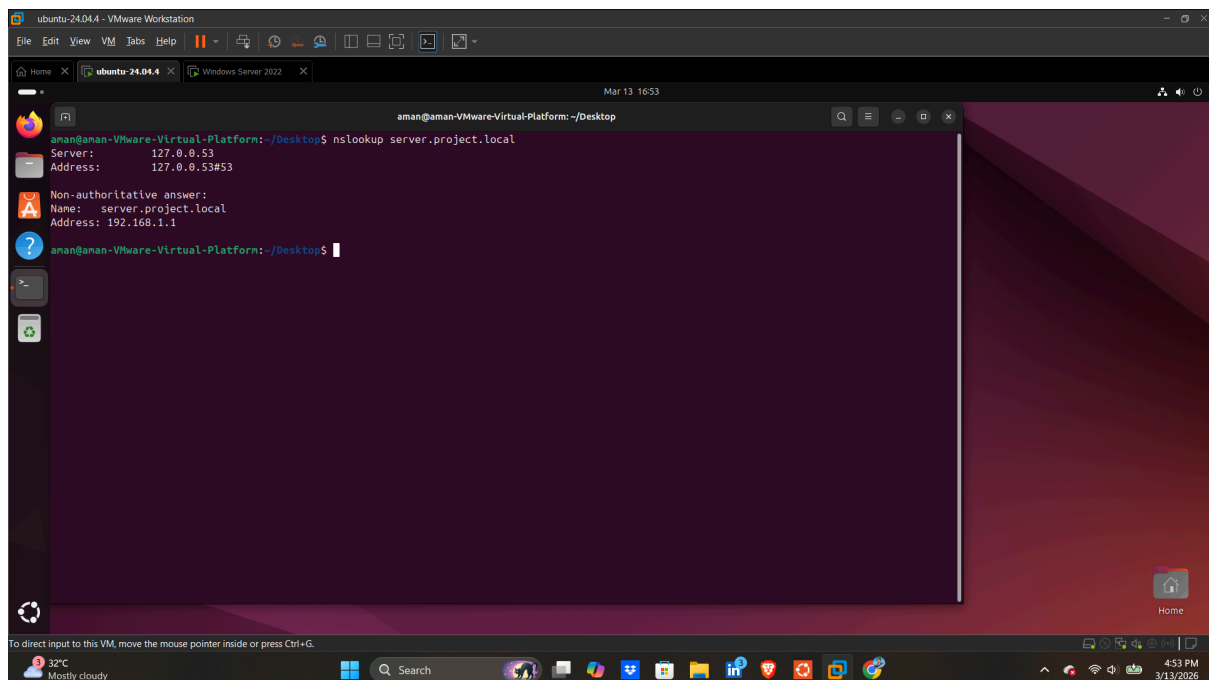
nslookup server.project.local

Purpose

This command verifies whether the **DNS server can resolve a domain name into an IP address**.

Example:

server.project.local → 192.168.1.1



```
ubuntu-24.04.4 - VMware Workstation
File Edit View VM Tabs Help
ubuntu-24.04.4 Windows Server 2022
Mar 13 16:53
aman@aman-VMware-Virtual-Platform:~/Desktop
aman@aman-VMware-Virtual-Platform:~/Desktop$ nslookup server.project.local
Server:
127.0.0.53
Address:
127.0.0.53#53
Non-authoritative answers:
Name: server.project.local
Address: 192.168.1.1
aman@aman-VMware-Virtual-Platform:~/Desktop$
```

Expected Output

Name: server.project.local

Address: 192.168.1.1

Meaning

- ✓ The **DNS server** is functioning correctly
 - ✓ The **Ubuntu client** is using the **Windows Server** as its **DNS server**
-

3 Route Test (Network Path Test)

Command:

```
tracert google.com
```

Purpose

This command shows the **path that data packets take from the client to the destination on the internet**.

It displays each **router (hop)** that the data passes through.

Example Output

```
1 192.168.1.1
2 ISP Router
3 Google Server
```

Meaning

The command traces the **network route step-by-step from the client to the destination server**.

Simple Summary

Test	Command	Purpose
Ping Test	ping 192.168.1.1	Checks if the server is reachable

DNS Test	nslookup server.project.local	Verifies domain name resolution
Route Test	tracert google.com	Checks the network path to the destination

Step 8: Understanding the DORA Process (DHCP Working Mechanism)

The **DORA process** explains how **DHCP** assigns an IP address to a **client device** automatically.

DORA stands for:

- **D → Discover**
- **O → Offer**
- **R → Request**
- **A → Acknowledge**

When a client device (such as an **Ubuntu machine**) needs an IP address, these four steps occur automatically between the **client and the DHCP server**.

1 Discover

When Ubuntu starts or connects to the network, the client sends a **broadcast message** to the network asking:

“Is there any DHCP server available that can assign me an IP address?”

This step is called **Discover**.

Example:

The **Ubuntu client** searches for a **DHCP server** on the network.

2 Offer

The **Windows Server (DHCP server)** responds to the client with an available IP address.

The server sends a message similar to:

“I have a free IP address available: 192.168.1.10.”

This step is called **Offer**.

Example:

The **DHCP server** offers an **IP address** to the Ubuntu client.

3 Request

After receiving the offer, the client responds by requesting that specific IP address.

The client sends a message indicating:

“I would like to use the IP address that you offered.”

This step is called **Request**.

Example:

The **Ubuntu client** requests the offered IP address from the server.

4 Acknowledge

Finally, the DHCP server confirms the assignment and officially allocates the IP address to the client.

The server sends a confirmation message stating:

“The IP address is now assigned to you.”

This step is called **Acknowledge**.

Example:

The **server** confirms and assigns the IP address to the client.

Simple Example (In Your Project Setup)

1. **Ubuntu Client** requests an IP address.
2. **Windows Server** offers **192.168.1.10**.
3. **Ubuntu Client** requests that IP address.
4. **Windows Server** confirms the assignment.

Final Result

Ubuntu Client IP Address = 192.168.1.10

Step 9: Troubleshooting (Project Demo)

Ubuntu Client Me Commands

(a) IP Address Check

Terminal me run karo:

ip a

Purpose:

Check karna ki Ubuntu ko DHCP se IP mila ya nahi.

Expected example:

```
inet 192.168.1.10/24
```

👉 Matlab DHCP server ne IP assign kar diya.

(b) Ping Test

```
ping 192.168.1.1
```

Purpose:

Check karna ki Ubuntu client **Windows Server tak reach kar raha hai ya nahi.**

Expected:

```
64 bytes from 192.168.1.1
```

👉 Matlab network connection working hai.

(c) DNS Test

```
nslookup server.project.local
```

Purpose:

Check karna ki **DNS server domain name ko IP me convert kar raha hai ya nahi.**

Expected:

```
Name: server.project.local
```

```
Address: 192.168.1.1
```

👉 Matlab DNS server correctly configured hai.

(d) Route Test

```
tracert 192.168.1.1
```

Purpose:

Check karna ki **data server tak kis route se ja raha hai.**

Example:

```
1 192.168.1.1
```

👉 Matlab Ubuntu directly server tak connect hai.

2 Windows Server Me Commands

Command Prompt open karo.

(a) IP Configuration

```
ipconfig
```

Purpose:

Check karna ki server ka IP address kya hai.

Expected:

```
IPv4 Address . . . . . : 192.168.1.1
```

(b) Ping Test

```
ping 192.168.1.1
```

Purpose:

Server ki network connectivity check karna.

Expected:

```
Reply from 192.168.1.1
```

(c) DNS Test

```
nslookup server.project.local
```

Purpose:

Check karna ki DNS server domain resolve kar raha hai.

Expected:

Name: server.project.local

Address: 192.168.1.1



Troubleshooting Summary

System	Command	Purpose
Ubuntu	<code>ip a</code>	IP address check
Ubuntu	<code>ping 192.168.1.1</code>	Server connectivity
Ubuntu	<code>nslookup server.project.local</code>	DNS resolution
Ubuntu	<code>tracert 192.168.1.1</code>	Network route
Windows	<code>ipconfig</code>	Server IP check
Windows	<code>ping</code>	Network connectivity
Windows	<code>nslookup</code>	DNS verification